

Targeting Toward Inferential Goals in Bayesian Item Response Models for Estimating Person-Specific Latent Traits

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ABSTRACT. When an item response model is fitted, a single set of scores is usually reported and used for every purpose: individual feedback, rankings, and statements about the group, such as the share of examinees below a benchmark. Two choices jointly determine what such reports contain, the prior placed on the latent trait distribution and the summary used to read scores off the posterior, and we evaluate them jointly against the three reporting goals in Bayesian Rasch and two-parameter logistic models, treating test reliability as an explicit design factor. In a large simulation the appropriate summary was determined by the reporting goal: posterior means performed best for individual scores, whereas the constrained Bayes and triple-goal estimators performed best for describing the group’s distribution, and ranking accuracy depended on test reliability alone. The appropriate prior was determined by the data regime: a flexible Dirichlet process mixture prior recovered non-normal trait distributions substantially better than the Gaussian default when tests were reliable and samples large, cost little when the population was in fact normal, but could worsen individual scores for clustered populations measured by short tests. Reliability governed the size of every gain. The comparison was then repeated for thirteen operational tests from the Item Response Warehouse; the same pattern reappeared, and the conventional default of a Gaussian prior read through posterior means carried up to twice the achievable distributional error. The whole workflow is implemented in the open-source R package DPMirt and two companion packages for prior elicitation and reliability-targeted data generation.

Keywords: Bayesian item response theory; person-specific latent traits; inferential goals; test reliability; Dirichlet process mixture; constrained Bayes; triple-goal estimator

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1. Introduction

An item response model fitted to a set of test responses is usually required to support more than one kind of report. In a curriculum-based assessment, for instance, the same calibration must provide an ability estimate for each examinee, a ranking that identifies the students most in need of intervention, and a statement about the group, such as the share of the class falling below a benchmark (Fuchs, 2004; Nelson et al., 2023). These are three different estimation problems, each with its own natural measure of error (Shen & Louis, 1998), and a single set of point estimates cannot in general serve all three well at once.

Standard practice nevertheless answers all three questions with one estimator. The latent trait distribution is assumed normal, and each person’s posterior mean, the expected a posteriori score of Bock and Mislevy (1982), is reported for every purpose. Both conventions have documented costs. When the true trait distribution departs from normality, a normal prior biases individual estimates and pulls the estimated distribution toward normality (Finch & Edwards, 2016; Paganin et al., 2023; Woods, 2006; Zhang et al., 2021). Even when the prior is correct, posterior means are less variable than the traits they estimate, because shrinkage removes part of the between-person variability that a distributional report is intended to describe (Louis, 1984; Mislevy et al., 1992); a histogram of reported scores therefore understates the heterogeneity of the group, and the shortfall grows as the test becomes less reliable.

Neither situation is unusual in applied data. In a recent census of 889 item-response data sets from the Item Response Warehouse (IRW; Domingue et al., 2025), between 30% and 52% of administrations fall below the conventional reliability threshold of .80, depending on whether reliability is defined leniently or strictly, and about one in ten fall below .50 under the strict definition (Lee, 2026e). In a companion analysis of 504 data sets, more than half showed estimated latent distributions that differ from the best-fitting normal by at least ten percentage points of cumulative mass somewhere along the trait scale, with skewed, heavy-tailed, flat, and occasionally two-humped shapes (Lee, 2026d). Low reliability and non-normal trait distributions therefore describe the conditions under which a large share of applied measurement operates.

Two remedies have been developed, largely in separate literatures. The first replaces the normal prior on the trait distribution with a flexible one, in particular a Dirichlet process mixture, so that the shape of the distribution is learned from the data (Duncan & MacEachern, 2008; Finch & Edwards, 2016; Paganin et al., 2023; San Martín et al.,

2011). The second keeps the model and changes the summary read off the posterior: the constrained Bayes estimator of Louis (1984) and Ghosh (1992) restores the spread that shrinkage removes, and the triple-goal estimator of Shen and Louis (1998) targets the group’s distribution and ranks directly. The flexible-prior literature judges its models almost entirely by the accuracy of individual estimates, while the summary literature has grown up around small-area estimation and multisite trials rather than measurement (Lee et al., 2025; Paddock et al., 2006). The two choices, however, act on the same reported scores: the prior determines the posterior, and the summary determines what is read off it. A flexible prior gains little if its fit is read through a mismatched summary, since the summary may discard the shape information that the prior recovered, and a suitable summary can compensate for a misspecified prior for some purposes but not for others. As far as we are aware, the two literatures have not been brought together in a systematic evaluation for item response models.

In this paper we evaluate the two levers jointly: three priors on the trait distribution (the Gaussian default and two Dirichlet process mixture priors that differ in how their concentration parameter is elicited) crossed with three posterior summaries (the posterior mean, the constrained Bayes estimator, and the triple-goal estimator), in both the Rasch and two-parameter logistic models, against all three reporting goals. Four features distinguish the evaluation from earlier comparisons. First, test reliability is an explicit design factor: the simulated data were generated by a method that solves the inverse problem of hitting a prespecified reliability (Lee, 2025), so estimator performance can be read along a designed ladder from .5 to .9, realized jointly by test length and discrimination rather than left to incidental test length. Second, the concentration parameter of the flexible prior is set by a design-conditional elicitation framework with known operating characteristics (Lee, 2026a) instead of an ad hoc default. Third, the simulation’s confirmatory claims were fixed in an analysis plan locked before the production runs, with multiplicity control and a calibration check on normal data. Fourth, the simulation is joined by thirteen real data sets from the IRW analyzed under the same pipeline, so the practical consequence of each choice can be shown on operational tests, and the whole workflow is available in the open-source R packages DPMirt, DPprior, and IRTsimrel (Lee, 2026b, 2026c, 2026f).

This study extends to the measurement domain the strategy that Lee et al. (2025) developed for multisite trials. That study already crossed Gaussian and Dirichlet process priors with the same three summaries, evaluated individual, rank, and distributional error, treated data informativeness as the governing moderator, and closed

with an empirical application; the question, the estimators, and the factorial logic all transfer. The model in which they must operate, however, does not. Site effects enter a Gaussian likelihood with known standard errors; person parameters enter a dichotomous likelihood jointly with unknown item parameters, so every posterior the summaries read is shaped by the item calibration as well. The latent metric has no units of its own, so its location, and under the two-parameter model its scale, exist only by convention; a finite test only partially identifies a freely estimated trait distribution, and no identification theorem covers the two-parameter case at all ([Section 2](#)). And the informativeness moderator is here a designed property of the instrument. Whether the multisite conclusions survive these conditions is the question this paper tests; the contribution is that preregistered transport test, together with the elicitation machinery, the operational-test evidence, and the software and diagnostics it produced, including a seed-stability rule for the triple-goal estimator on short forms.

The results can be summarized as follows. The choice of summary was decided by the goal: distribution-matched summaries performed best for distributional questions and the posterior mean for individual scores, while ranking accuracy was determined by the instrument’s reliability and by nothing else we varied. The choice of prior was decided by the data regime: the flexible prior improved distribution recovery where the population departed from normality and the data were informative enough to show it, at a small certified cost, about 2% on average, where the population was in fact normal. The instrument’s position on the reliability ladder governed the size of the achievable gains: below tier .7 most of the gain came from the summary, while from .8 upward the two choices compounded. The same structure reappeared in the real data, where the conventional default carried 1.2 to 2 times the distributional error of better-matched choices.

The plan of the paper is as follows. [Section 2](#) sets out the Bayesian Rasch and two-parameter models, reliability, and identification. [Section 3](#) defines the three inferential goals and the three posterior summaries, and [Section 4](#) develops the flexible prior and its elicitation. [Sections 5](#) and [6](#) describe the design and results of the simulation, and [Section 7](#) takes the findings to real tests from the IRW. [Section 8](#) distills practical guidance and introduces the software; a worked example is given in [Appendix G](#). We close with a discussion in [Section 9](#).

2. Bayesian Item Response Models for Person Parameters

2.1. The Rasch and Two-Parameter Logistic Models. Let $U_{pi} \in \{0, 1\}$ denote the response of person $p = 1, \dots, P$ to item $i = 1, \dots, I$, with realization u_{pi} , and let $\mathbf{u}_p = (u_{p1}, \dots, u_{pI})^\top$ collect person p 's responses. Under the two-parameter logistic (2PL) model, the responses are conditionally independent Bernoulli variables with success probability π_{pi} given by

$$\text{logit}(\pi_{pi}) = \lambda_i(\theta_p - \beta_i), \quad (1)$$

where θ_p is the latent trait of person p , β_i is the difficulty of item i , and $\lambda_i > 0$ is its discrimination. The Rasch model is the special case $\lambda_i \equiv 1$ (Debelak et al., 2022; Rasch, 1960). Under the Rasch model the person total $r_p = \sum_i u_{pi}$ is sufficient for θ_p given the item parameters; under the 2PL the corresponding weighted total $\sum_i \lambda_i u_{pi}$ is sufficient only when the discriminations are known.

When the item parameters are treated as known, or replaced by calibrated estimates in the usual plug-in fashion (Baker & Kim, 2004), the trait can be estimated from $\mathbf{u}_p$ alone by maximum likelihood, giving $\hat{\theta}_p$ as the root of $\sum_i \lambda_i (u_{pi} - \pi_{pi}) = 0$, or by the weighted likelihood correction of Warm (1989), which removes the leading bias term and yields finite estimates for perfect and zero scores. The sampling precision of such within-person estimates is governed by the Fisher information

$$\mathcal{J}(\theta_p) = \sum_{i=1}^I \lambda_i^2 \pi_{pi}(1 - \pi_{pi}), \quad (2)$$

and the squared standard error of $\hat{\theta}_p$ is $\text{se}(\hat{\theta}_p)^2 = 1/\mathcal{J}(\theta_p)$ to the usual asymptotic order. Information is largest for persons located near the item difficulties and, under the 2PL, rises with the squared discriminations, so a short or poorly targeted test measures some regions of the trait scale far better than others.

2.2. Test Reliability. Reliability summarizes how the measurement error just described compares with the spread of the traits being measured. Because several non-equivalent coefficients circulate under the one name (Adams, 2005; Lee, 2026e; Sijtsma, 2009), we state ours exactly. Writing σ_θ^2 for the variance of θ under its population distribution G and $\text{MSEM} = \mathbb{E}_G\{1/\mathcal{J}(\theta)\}$ for the population-average error variance, the design coefficient used throughout this paper is

$$\bar{\rho} = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \text{MSEM}}. \quad (3)$$

This is the population analogue of the person-separation reliability of the Rasch tradition (Wright & Masters, 1982): the proportion of the variance of unshrunk trait estimates that reflects true between-person differences rather than measurement error. Two related coefficients should be distinguished from it. The marginal reliability of maximum likelihood scores (Andersson & Xin, 2018; Green et al., 1984) averages the ratio itself, $\mathbb{E}_G\{\sigma_\theta^2 \mathcal{J}(\theta)/(\sigma_\theta^2 \mathcal{J}(\theta) + 1)\}$, and the average-information form replaces MSE in Equation (3) by $1/\mathbb{E}_G\{\mathcal{J}(\theta)\}$; by Jensen’s inequality the three are ordered, with Equation (3) the strictest, and we use Equation (3) throughout. It is also stricter than EAP-based empirical reliability, which credits the estimator for the very shrinkage whose consequences this paper studies; across 889 real data sets the lenient and strict definitions differ by a median of about .045 and classify 30% against 50% of the corpus as falling below .80 (Lee, 2026e). Where we report a reliability, it is Equation (3) unless stated otherwise.

2.3. The Hierarchical Bayesian Formulation. The plug-in account treats the calibrated item parameters as known and each person in isolation. A hierarchical Bayesian formulation removes both restrictions: it propagates item-calibration uncertainty into the person estimates (Fox, 2010; Patz & Junker, 1999), and it is the natural setting in which the population distribution G can itself be modeled flexibly, which is the route we take in Section 4. The model adds to Equation (1) a population level for the traits,

$$\theta_p \mid G \stackrel{\text{iid}}{\sim} G, \quad p = 1, \dots, P, \quad (4)$$

with the conventional parametric choice $G = N(\mu_\theta, \sigma_\theta^2)$ and hyperparameters $\delta = (\mu_\theta, \sigma_\theta^2)^\top$, together with priors on the item parameters and on δ ; the exact specifications fitted in this paper, including the sum-to-zero item prior and the hyperpriors, are given in Appendix A. We refer to these as levels of one joint model rather than as estimation stages, since all parameters are updated jointly.

The conditional posterior of a single trait given the item parameters and δ is

$$p(\theta_p \mid \mathbf{u}_p; \boldsymbol{\beta}, \boldsymbol{\lambda}, \delta) \propto p(\mathbf{u}_p \mid \theta_p, \boldsymbol{\beta}, \boldsymbol{\lambda}) p(\theta_p; \delta), \quad (5)$$

and under a normal approximation to the likelihood contribution its mean takes the familiar precision-weighted form

$$\mathbb{E}(\theta_p \mid \mathbf{u}_p; \boldsymbol{\beta}, \boldsymbol{\lambda}, \delta) \approx w_p \hat{\theta}_p + (1 - w_p) \mu_\theta, \quad w_p = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \text{se}(\hat{\theta}_p)^2}. \quad (6)$$

Here w_p is the shrinkage weight: persons measured precisely keep their data-based estimate, and persons measured poorly are pulled toward the population mean. Averaged over persons, the weights are governed by the reliability in Equation (3), so the lower the reliability, the more the ensemble of posterior means contracts around μ_θ . This contraction is optimal for estimating each θ_p separately and is what distorts distributional summaries, a tension we take up in Section 3. (What “the variance of $\hat{\theta}_p$ ” means depends on the estimation framework; in the joint Bayesian model the relevant quantity is the posterior variance $v_p = \text{Var}(\theta_p \mid \mathbf{u})$, which Section 3 uses throughout.)

2.4. Identification. Because the latent metric has no units of its own, the models above are not identified without conventions, and which conventions are available depends on the model (San Martín, 2016). Under the Rasch model, adding a constant to every θ_p and every β_i leaves all response probabilities unchanged, so one location restriction is required; the unit discrimination fixes the scale. Under the 2PL the transformation $\theta^* = a(\theta - c)$, $\beta^* = a(\beta - c)$, $\lambda^* = \lambda/a$ leaves Equation (1) invariant for any $a > 0$ and c , so both a location and a scale restriction are required.

The choice of restriction is not innocuous when G is an estimand. The common statement that “ $\theta_p \sim N(0, 1)$ fixes the location and scale” conflates two different operations: in a random-effects Rasch model whose G is known up to location and scale, setting $\sigma_\theta^2 = 1$ restricts a parameter that the data identify, which can misspecify the model, rather than removing an indeterminacy (San Martín, 2016). We instead identify the location through the items, constraining $\sum_i \beta_i = 0$, which leaves μ_θ free and estimable; under the 2PL our default adds a unit geometric-mean constraint on the discriminations, applied either in-model or by post-hoc rescaling of the posterior draws, as in Paganin et al. (2023). When G carries a nonparametric prior the issue becomes more delicate: centering the prior on G does not center G itself, and a finite test identifies only finitely many functionals of G , so part of what a fitted flexible distribution reports is convention. We return to this in Section 4, and Appendix B collects the formal statements.

3. Three Inferential Goals and Goal-Specific Posterior Summaries

3.1. Goals, Targets, and Losses. Following Shen and Louis (1998), we distinguish three inferential goals for the person parameters, summarized in Table 1. The first is to estimate each individual trait θ_p ; the natural loss is the average squared error $\text{MSEL} = P^{-1} \sum_p (a_p - \theta_p)^2$ for candidate estimates a_p , and the Bayes rule under this

loss is the posterior mean, $\eta_p = \mathbb{E}(\theta_p \mid \mathbf{u})$, which in the measurement literature is the EAP score. The second goal is to rank the persons. The target is the vector of true ranks $R_p = \sum_q I\{\theta_q \leq \theta_p\}$, where the indicator $I\{A\}$ equals 1 when A holds and 0 otherwise, and we score rank estimates A_p by the squared error of the corresponding percentile ranks, $\text{MSELP} = P^{-1} \sum_p \{(A_p - R_p)/P\}^2$, which divides by P^2 so that the loss is comparable across group sizes (Lee et al., 2025). Ranks estimated from noisy measurements carry substantial uncertainty of their own (Goldstein & Spiegelhalter, 1996).

TABLE 1. Three Inferential Goals for Person-Specific Latent Traits

Goal	Target	Loss function	Matched summary
1. Estimate individual traits	True θ_p , $p = 1, \dots, P$	$\text{MSEL} = P^{-1} \sum_p (a_p - \theta_p)^2$	Posterior mean (PM)
2. Rank individuals	True ranks $R_p = \sum_q I\{\theta_q \leq \theta_p\}$	$\text{MSELP} = P^{-1} \sum_p \{(A_p - R_p)/P\}^2$	Posterior expected ranks; in practice information-limited
3. Estimate the empirical distribution	Realized EDF $G_P(t) = P^{-1} \sum_p I\{\theta_p \leq t\}$	$D_{\text{KS}} = \sup_t A(t) - G_P(t) $	Triple-goal (GR); CB restores spread only

Note. a_p , A_p , and $A(t)$ denote a candidate estimator of the trait, its induced rank, and the EDF of the reported estimates, respectively; $I\{A\}$ is the indicator of the event A . G_P is the empirical distribution of the P realized abilities in the sample, not the population distribution G ; Section 3 develops the distinction. The theoretical development of Goal 3 uses integrated squared error loss on the CDF, under which the GR construction is optimal (Shen & Louis, 1998); the KS distance shown is the prespecified empirical criterion for the same target. Example questions: What is examinee p 's location on the latent scale (Goal 1)? Which ten examinees scored lowest (Goal 2)? What share of the group falls below a cutoff (Goal 3)? PM = posterior mean; CB = constrained Bayes; GR = triple-goal estimator; EDF = empirical distribution function; KS = Kolmogorov–Smirnov distance.

The third goal is to estimate the distribution of the traits in the group at hand. The target is the empirical distribution function (EDF) of the P realized abilities,

$$G_P(t) = \frac{1}{P} \sum_{p=1}^P I\{\theta_p \leq t\}, \quad (7)$$

and we measure the discrepancy between a candidate distribution estimate $A(t)$ and the target by the Kolmogorov–Smirnov distance $D_{\text{KS}} = \sup_t |A(t) - G_P(t)|$. One

distinction made here matters throughout the paper: the decision theory of Shen and Louis (1998) develops Goal 3 under integrated squared error loss on the CDF (ISEL), for which the Bayes action has a closed form, whereas D_{KS} is the prespecified empirical criterion we evaluate against, chosen for its reading as a worst-case share discrepancy; no optimality theorem attaches to D_{KS} itself. Two clarifications prevent possible confusion. First, G_P is the distribution of this group’s realized traits, not the population distribution G of Equation (4); with P persons the group’s own EDF is the quantity a class report or a below-benchmark share actually describes. Second, the posterior expectation of G_P at a point t ,

$$\bar{G}_P(t) = \frac{1}{P} \sum_{p=1}^P \mathbb{P}(\theta_p \leq t \mid \mathbf{u}), \quad (8)$$

is an average of posterior probabilities and therefore a number between 0 and 1; it is neither a rank estimate nor a trait estimate, although trait values will be read off its inverse below.

3.2. Posterior Mean, Constrained Bayes, and Triple-Goal Summaries. The posterior mean solves Goal 1 by construction, and its weakness for Goal 3 follows from the shrinkage identity Equation (6) (Louis, 1984; Mislevy et al., 1992): the variance of the η_p across persons understates the posterior estimate of $\text{Var}(\theta)$ by the average posterior variance $\bar{v} = P^{-1} \sum_p v_p$, so the EDF of the posterior means is too narrow, and increasingly so as reliability falls. For ranks the situation is different again. The Bayes rule under squared rank loss is the posterior expected rank, discretized to integers; but under the Rasch model, in which all persons answer the same items, the posteriors are stochastically ordered in the total score, so every monotone summary induces the same ordering (Laird & Louis, 1989; Shen & Louis, 1998), and rank accuracy is limited by the information in the data rather than by the choice of summary.

The constrained Bayes (CB) estimator repairs the variance shortfall directly. In the posterior-independence form implemented here (Ghosh, 1992), it is the affine transformation

$$\theta_p^{\text{CB}} = \bar{\eta} + a(\eta_p - \bar{\eta}), \quad a = \left(1 + \frac{(1 - 1/P)\bar{v}}{V_\eta}\right)^{1/2}, \quad (9)$$

where $\bar{\eta}$ is the mean of the posterior means and $V_\eta = P^{-1} \sum_p (\eta_p - \bar{\eta})^2$ their variance with divisor P (with the finite-sample factor written out, the expression coincides with the familiar form that uses the divisor- $(P - 1)$ sample variance); a is undefined

in the degenerate case $V_\eta = 0$, in which all posterior means coincide. Since $a > 1$, CB is an inflation of the posterior means about their center, chosen so that the reported ensemble matches the posterior estimate of the group mean and variance. Being affine, CB preserves the ordering of the posterior means exactly and inherits their distributional shape: it can restore spread, but it cannot restore a valley between two modes that shrinkage has filled in. A scale correction is nevertheless worthwhile, and CB requires nothing beyond the posterior means and variances.

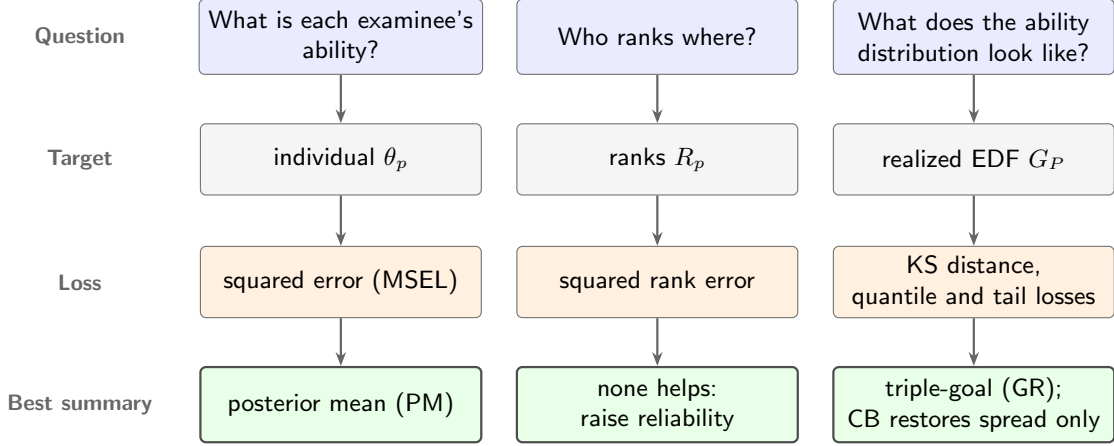
The triple-goal (GR) estimator of Shen and Louis (1998) rebuilds the reported ensemble from the estimated distribution and the estimated ranks. Let $\hat{R}_p$ denote person p 's discretized posterior expected rank, and define P equally spaced quantiles of $\bar{G}_P$ by

$$\hat{U}_j = \bar{G}_P^{-1}\left(\frac{2j-1}{2P}\right), \quad j = 1, \dots, P, \quad \theta_p^{\text{GR}} = \hat{U}_{\hat{R}_p}. \quad (10)$$

The first step discretizes $\bar{G}_P$ into the P -point distribution that is optimal under integrated squared error loss (ISEL); the second assigns those support points to persons in rank order. Because the assignment is a permutation, the EDF of the GR estimates equals the discretized $\bar{G}_P$ exactly, which is the property CB cannot deliver; the cost falls on Goal 1, where GR's squared-error regret relative to PM is the mean squared gap between the support points, assigned at the estimated ranks, and the posterior means (the ordered-gap form when the two rankings agree).

A single vector of point estimates cannot in general attain the optima for individual error and for distribution recovery at once when the posteriors overlap, since the ISEL-optimal support points differ from the posterior means (Shen & Louis, 1998). The choice among PM, CB, and GR is therefore a decision about which loss the report will be scored by, and this is the theoretical reason the prior and the summary must be studied jointly: the prior determines the posterior on which every summary operates, while the summary determines which features of that posterior survive into the report. Figure 1 summarizes the resulting mapping from questions to targets, losses, and summaries; the empirical content of its last row, including the rank column, whose accuracy turns out to be set by the instrument rather than by any summary, is established in Section 6.

FIGURE 1. From Reporting Questions to Targets, Losses, and Matched Summaries



Note. Each column is one of the three inferential goals of Table 1. The loss row lists the empirical criteria evaluated in this paper; the theoretical loss under which the GR construction is derived is the integrated squared error on the CDF (Section 3.1). The last row states the summary that the theory matches to each goal, with the rank column reflecting the information limit established empirically in Section 6. PM = posterior mean; CB = constrained Bayes; GR = triple-goal; EDF = empirical distribution function; MSEL = mean squared error loss; KS = Kolmogorov–Smirnov distance.

4. Flexible Priors Through Dirichlet Process Mixtures

4.1. The Dirichlet Process. The Dirichlet process (DP) of Ferguson (1973) is a prior on distributions. Its parameter is a finite measure $\alpha(\cdot)$ on the real line, which it is convenient to split into a scalar total mass and a probability measure: the total mass $\alpha \equiv \alpha(\mathbb{R})$ is the concentration parameter, and the normalized measure G_0 is the base measure, which is also the prior mean of the process in the sense that $\mathbb{E}\{F(A)\} = G_0(A)$ for every set A . A draw F from $\text{DP}(\alpha, G_0)$ is almost surely discrete, as the stick-breaking construction of Sethuraman (1994) makes visible: F concentrates its mass on countably many atoms drawn from G_0 , with weights $w_1 = V_1$, $w_h = V_h \prod_{m < h} (1 - V_m)$ for independent $V_h \sim \text{Beta}(1, \alpha)$. Small α puts most of the stick-breaking mass on few atoms; large α spreads it thinly, and the process concentrates around G_0 .

If the random measure is integrated out, the sequence of draws from F follows the Pólya-urn scheme of Blackwell and MacQueen (1973): the $(p+1)$ th draw repeats one of the first p values with probability proportional to its multiplicity, and takes a fresh value from G_0 with probability proportional to α . The induced random partition of

persons into groups sharing a value is what is popularly described as the Chinese restaurant process; it is not a separate model but the marginal partition law of the DP, and we introduce it in that order because the clustering behavior that matters for elicitation, in particular the distribution of the number of occupied clusters K_P among P persons, is a property of this marginal law (Antoniak, 1974). The expected number of clusters grows only logarithmically in P , at rate governed by α , so the DP is a parsimony prior: it entertains new clusters reluctantly and only when the data support them.

4.2. A DP Mixture Prior for the Trait Distribution. Because a draw from the DP is discrete, it is not used directly as a prior for a continuous trait distribution. Following Lo (1984), we place the DP one level up, letting G be a countable mixture of normal kernels whose parameters are drawn from the process, so that

$$\theta_p \mid G \stackrel{\text{iid}}{\sim} G, \quad G(\cdot) = \int N(\cdot \mid \tilde{\mu}, \tilde{\sigma}^2) dF(\tilde{\mu}, \tilde{\sigma}^2), \quad F \sim \text{DP}(\alpha, H_0), \quad (11)$$

with base measure $H_0 = N(0, \sigma_\mu^2) \times \text{InvGamma}(\nu_1, \nu_2)$ on the kernel means and variances. This replaces the level-2 distribution in Equation (4) and leaves the rest of the hierarchical model of Section 2.3 unchanged; it is the construction that Duncan and MacEachern (2008) introduced to item response modeling, that San Martín et al. (2011) generalized and analyzed, and that Paganin et al. (2023) made computationally routine. (Miyazaki and Hoshino (2009) used the same machinery on the other side of the model, freeing the shape of the item response functions rather than of the trait distribution.) The number of mixture components, their locations, and their scales are all inferred; with the base measure centered at zero, the prior mean of G is a symmetric and unimodal scale mixture of normals: it shares the Gaussian reference’s center, symmetry, and unimodality, though it is heavier-tailed than any single normal, and individual draws from the process depart from it a priori. In this sense the flexible prior is a safeguard rather than a rival specification: it reproduces the conventional model’s broad shape in expectation while permitting the data to move the fitted distribution away from it.

Two identification points from Section 2.4 sharpen under Equation (11). First, centering H_0 at zero centers the prior expectation of G , not G itself; a realized draw has a nonzero mean almost surely, so the constraint that identifies the location must act on the item side, and our fits accordingly retain $\sum_i \beta_i = 0$ with G left entirely free (San Martín et al., 2011). Second, a finite test identifies only a finite set of functionals of G , of the order of the number of items, so the fitted shape of G combines data

information with prior structure in proportions that worsen as tests shorten (San Martín et al., 2011). Both points argue for treating the concentration hyperprior as a design decision rather than a nuisance default, which is the subject we turn to next; the formal statements, including the open status of semiparametric identification under the 2PL, are collected in [Appendix B](#).

4.3. Eliciting the Concentration Hyperprior by Design. The data are typically weakly informative about α , so its hyperprior, conventionally $\alpha \sim \text{Gamma}(a, b)$ (Escobar & West, 1995), materially shapes the posterior clustering (Dorazio, 2009; Murugiah & Sweeting, 2012). Off-the-shelf choices can behave poorly in ways that are not obvious. A diffuse Gamma prior can induce posterior collapse onto one or two clusters even when the true structure is richer, and a prior tuned only to the expected number of clusters can still concentrate most of the stick on a single dominant component, the phenomenon the elicitation framework below terms the unintended prior, and which Vicentini and Jermyn (2025) raise through the prior distribution of the stick-breaking weights: matching $\mathbb{E}(K_P)$ says nothing about the implied mass of a dominant component. The framework monitors this through the first stick-breaking weight w_1 , whose size-biased interpretation is the asymptotic mass of the cluster containing a randomly chosen person; w_1 is not necessarily the largest component, so its tail probabilities bound dominance from below.

We therefore set the hyperprior by the design-conditional elicitation framework of Lee (2026a), implemented in the DPprior package (Lee, 2026c). The analyst states a target mean and variance for the number of occupied clusters K_P at the design’s own P , and a two-stage moment-matching procedure (a closed-form negative-binomial initializer followed by Newton refinement against the exact finite- P moments of the Antoniak distribution) returns the $\text{Gamma}(a, b)$ hyperparameters that reproduce those targets exactly. Because the first stick-breaking weight satisfies $\mathbb{P}(w_1 > t \mid a, b) = \{b/(b - \log(1 - t))\}^a$ in closed form, the implied dominance behavior is checked as part of the same procedure, and a dual-anchor variant can trade cluster-count fidelity against a stated bound on $\mathbb{P}(w_1 > t)$ when the two conflict.

The simulation and the case studies both carry two elicitation arms built this way. The focused arm encodes the belief that a trait distribution is adequately described by a handful of occupied clusters at the design’s sample size, stated with medium confidence; the broad arm roughly doubles the cluster-count target and widens its variance, and provides a sensitivity contrast. The exact targets, the resulting Gamma hyperparameters at each N , and their induced K_P and w_1 diagnostics are tabulated

in [Appendix C](#). The comparison of the two arms provides a direct check on how much the elicitation matters once the data are informative; the answer, reported in [Section 6](#), is that it matters little.

5. Simulation Design

5.1. Factors and Replication. The simulation crosses two item response models (Rasch, 2PL), three latent shapes, five target reliabilities (.5 to .9), and four sample sizes ($N = 50$ to 500), giving 120 conditions; the factors and their levels are collected in Panel A of [Table 2](#). Each condition was replicated with five independently generated item banks, and each bank with four response draws, so that 20 data sets underlie every condition (2,400 data sets in total) and item-bank variability is propagated into the reported uncertainty rather than held fixed by the design. Every data set was analyzed under all three priors on G (Gaussian, DP focused, DP broad), and all three summaries (PM, CB, GR) were read from each fit, so all comparisons are replicate-paired: 7,200 fits and 115,200 loss records in all. We stopped at $N = 500$ because earlier exploratory builds indicated that the qualitative map had stabilized, and reallocated the computing budget to replication.

The three latent shapes were standardized to mean zero and unit variance. The normal shape is the calibration control: a method that is intended to help under non-normality must first be shown to cost little when the population is in fact normal. The skewed shape (skewness constant 2.0) and the bimodal shape (mode separation 0.90, two well-separated modes) represent the two departure families that dominate the empirical shape census of Lee ([2026d](#)), in which skewness was the most common departure and pronounced multimodality a minority feature; the bimodal condition is thus a stress test rather than a modal case.

5.2. Reliability-Targeted Data Generation. Treating reliability as a design factor requires generating data whose reliability hits a prespecified target, which is an inverse design problem: no closed form maps test length and item parameters to a desired $\bar{\rho}$. We used the reliability-targeted simulation framework of Lee ([2025](#)), implemented in IRTsimrel (Lee, [2026f](#)), which separates the structure of the data-generating process from its scale and calibrates a single global discrimination multiplier c^* , so that item discriminations $\lambda_i = c^* \lambda_{i,0}$ attain the target value of the design coefficient in [Equation \(3\)](#). Strictly nested item subsets of one balanced bank template per model family form the coarse ladder from tier to tier, and c^* was then calibrated separately in every model-by-bank-by-tier-by-shape cell (150 calibration cells) on a refinement

TABLE 2. Simulation Design and the Realized Reliability Ladder

Panel A: Data-generating and analytic factors

Factor	Levels
Item response model	Rasch; 2PL (crossed, co-primary)
Latent shape	normal (control); skewed; bimodal (constants in the note)
Target reliability $\bar{\rho}$	.5, .6, .7, .8, .9
Sample size N	50, 100, 200, 500
Fitted prior on G	Gaussian; DP focused; DP broad (same data)
Posterior summary	PM; CB; GR (same fit)
Replication	5 item banks $\times$ 4 response draws = 20 per condition

Panel B: Realized ladder per model and tier

Target $\bar{\rho}$	Rasch			2PL		
	Achieved	Items	c^*	Achieved	Items	c^*
.5	.4999	8.0	0.946	.4999	7.4	0.966
.6	.6001	12.0	0.910	.6001	10.8	0.885
.7	.6997	18.0	0.924	.7000	16.4	0.887
.8	.8002	31.0	0.916	.7999	28.0	0.919
.9	.9000	68.2	0.918	.9001	62.2	0.927

Note. All three latent shapes are standardized to mean zero and unit variance; the skewed shape fixes its skewness constant at 2.0 and the bimodal shape its mode separation at 0.90. The full grid crosses 2 models $\times$ 3 shapes $\times$ 5 reliability tiers $\times$ 4 sample sizes = 120 conditions, each with 20 replicate data sets (2,400 data sets; 7,200 fits; 115,200 loss records). Panel B reports, per model and tier, the achieved design coefficient, the mean number of items over the five banks, and the mean calibrated global discrimination multiplier c^* . The achieved coefficient is computed analytically at the calibrated multiplier rather than estimated from the generated responses, and it matches its target within .003 everywhere (largest absolute error .0027). Item counts and c^* move together along the ladder, so reliability tiers should be read as a joint length-and-discrimination instrument rather than a manipulation of reliability alone (Section 5). DP = Dirichlet process; 2PL = two-parameter logistic model; PM = posterior mean; CB = constrained Bayes; GR = triple-goal estimator.

grid and certified on an independent population grid. Panel B of Table 2 shows the realized ladder: the achieved coefficient matches its target within .003 everywhere, with mean test lengths running from about 7–8 items at $\bar{\rho} = .5$ to 62–68 items at $\bar{\rho} = .9$; across the underlying 150 calibration cells the frozen multipliers span 0.79 to 1.09 (tier means 0.89 to 0.97 in Panel B). The bank templates fix a balanced difficulty layout that preserves coverage of the standardized ability range as the nested subsets grow, rather than drawing difficulties at random, and the same layout is shared across shapes within a tier; the empirical difficulty pools available in IRTsimrel (Lee, 2025) offer an alternative anchored in operational banks, and the realism burden that

random $N(0, 1)$ difficulties cannot carry is borne here by the real-data case studies of [Section 7](#).

One property of this construction must be kept in view when reading every result that follows. Reliability, test length, and discrimination move together along the realized ladder (the correlation between achieved reliability and log item count across cells is .98), so the reliability factor is a joint length-and-discrimination instrument, and effects of $\bar{\rho}$ are effects of that bundle; the design does not identify a causal partition among its components, and matching the scalar coefficient does not equalize the full information curves across shapes. We return to this limit in [Section 9](#).

5.3. Estimation and Loss Evaluation. All models were fitted by Markov chain Monte Carlo in NIMBLE (de Valpine et al., [2017](#)) through the DPMirt package, with four chains per fit and sampler budgets fixed in advance (8,000 to 40,000 iterations depending on model and prior; the 2PL budgets were raised once, before production, after a pilot mixing failure). The DP mixture was sampled in its marginal partition representation (Neal, [2000](#)), following the computational strategy of Paganin et al. ([2023](#)), with a truncation guard that was reached in 5 of 4,800 DP fits. Convergence screening excluded nothing: all 7,200 fits completed, none crossed the hard failure criteria, and 2.6% carried soft warnings; diagnostics are tabulated in [Appendix D](#). There are no separate frequentist arms, since the Gaussian-prior posterior mean is operationally the EAP default that applied work reports.

Losses were computed against each replication’s realized traits, on a common scale anchored to the data-generating parameters, for five loss families: individual squared error (MSEL), squared percentile-rank error (MSELP), the KS distance of [Section 3.1](#) (the primary distributional loss), a quantile loss on the nine deciles, and a tail-share loss at four cutoffs. The tail family proved uninformative at this scale, because a prespecified ratio floor binds for 60% to 85% of comparator values, and we accordingly make no tail claims from the simulation.

5.4. Hypotheses and Analysis Plan. The analysis plan was locked before the production fits. It registers five primary hypotheses, tested as one family with Holm’s correction at $\alpha = .05$: the flexible pipeline improves distribution recovery under non-normality (H1); the advantage grows with sample size (H2); it differs across model families and shapes (H3); it does not harm individual accuracy where non-normality meets $N \geq 200$, tested as non-inferiority within a 5% margin (H4); and the focused

elicitation beats the broad one (H5). Two secondary hypotheses, registered by amendment before any production fit, carry the paper’s central questions: the advantage steepens along the reliability ladder (H6), and the summary that minimizes loss depends on the inferential goal (H7). Alongside the hypothesis families, a calibration control had to show that the flexible prior wins nothing on normal data before any claim could proceed, and every cell was screened for individual-accuracy harm at prespecified thresholds. One registration event followed the results: a rule requiring the flagged safety cells to be disclosed wherever the H4 average is stated, which [Section 6](#) obeys. The estimation machinery, the per-cell evidence rules, the two ratio conventions used in reporting, and the complete registration timeline are given in [Appendix D](#).

5.5. Transparency and Openness. We report how the simulation’s conditions were determined, all manipulated factors, and all prespecified hypotheses; no fitted replication was excluded from analysis. The analysis plan was locked and hash-fingerprinted before the production fits but was not deposited in an external registry. The case studies in [Section 7](#) are observational applications to public data sets from the Item Response Warehouse (Domingue et al., [2025](#)) and support no causal claims (Appelbaum et al., [2018](#)). Software versions, sampler budgets, seeds, and archives are recorded in [Appendix A](#), and the packages are publicly available ([Section 8](#)).

6. Simulation Results

[Table 3](#) collects the confirmatory estimates: all five primary hypotheses and both secondary hypotheses were supported. Before the hypotheses are read, the calibration control must be examined, since any recommendation to fit a flexible prior should first show what the flexibility costs when the conventional model is correct. The cost was small. In the 40 normal-population cells the flexible pipeline won the registered distribution-recovery comparison in none (0 of 40, against a chance ceiling of .05), the estimated average cost of carrying the DP machinery on normal data was 1.6% in KS loss (log ratio +0.016, 95% CI [0.002, 0.030]), and realized type-I rates across the hypothesis family, computed from the post-hoc operating-characteristic analysis, were .035 to .048 against the nominal .05. Three individual normal cells did resolve a real flexible-prior cost, the largest 32% at the 2PL, $\bar{\rho} = .5$, $N = 500$ corner; this appears to be the cost of forcing a clustering prior onto abundant data that contain no clusters. All claims that follow rest on this control. The subsections take the results in the order of the paper’s argument: the summary lever first, then the prior

lever and its dependence on reliability, the model-family contrast, individual accuracy with its one systematic exception, and finally the two comparisons that produced null results.

TABLE 3. Confirmatory and Secondary Results From the Locked Analysis Plan

Hypothesis	Estimate	95% CI	p_{Holm}	Decision
<i>Primary family (Holm over five)</i>				
H1: flexible prior improves distribution recovery (non-normal cells)	-0.221	[-0.246, -0.195]	1.4×10^{-28}	supported
H2: the advantage grows with sample size (per doubling of N)	-0.114	[-0.123, -0.106]	1.2×10^{-131}	supported
H3: model-by-shape dissociation (Rasch-bimodal; 2PL-skew)	-0.127	[-0.175, -0.078]	1.0×10^{-4}	supported
H4: no individual-accuracy harm in the opportunity region (non-inferiority)	-0.070	[-0.080, -0.060]	2.3×10^{-23}	supported ^a
H5: focused elicitation beats broad	-0.011	[-0.017, -0.005]	1.0×10^{-4}	supported ^b
<i>Secondary family (Holm over two)</i>				
H6: advantage steepens along the reliability ladder (slope on $\bar{\rho}$)	-0.514	[-0.587, -0.441]	4.4×10^{-41}	supported
H7: summary-by-goal reversal (difference-in-differences) ^c	+0.281	[+0.210, +0.351]	8.6×10^{-6}	supported
<i>Gates and diagnostics (outside the families)</i>				
Calibration control: flexible-prior wins on normal cells	+0.016	[+0.002, +0.030]	—	pass (0/40 wins)
H6 diagnostic: the same slope within normal cells	-0.029	[-0.131, +0.074]	—	quiet

Note. Estimates are means of replicate-paired log loss ratios (or slopes of those ratios) from a linear mixed model with a form-family random intercept; negative values favor the flexible pipeline. The plan was locked before the production fits; its SHA-256 fingerprint travels with every confirmatory record. ^aH4 holds on average but not uniformly: seven bimodal cells at the two lowest reliability tiers show individual-accuracy costs with intervals excluding parity (2PL/500/.5 1.138 [1.115, 1.162]; 2PL/200/.5 1.122 [1.095, 1.149]; Rasch/200/.5 1.094 [1.048, 1.136]; 2PL/200/.6 1.092 [1.070, 1.116]; Rasch/500/.5 1.076 [1.059, 1.093]; 2PL/500/.6 1.072 [1.059, 1.084]; Rasch/200/.6 1.052 [1.030, 1.077]; Table 4). ^bThe H5 magnitude is specification-dependent; under the strict convergence-pass sensitivity it attenuates to -0.003 with a two-sided CI spanning zero (Appendix D). ^cH7 is the PM-to-GR change in log loss on the registered distributional composite minus the same change on individual squared error; its components are unequal (GR 21% worse on squared error; PM about 9% worse on the composite, which includes the floor-bound tail family), and the estimate licenses the reversal claim, not a cost of either summary alone (Section 6.1). CI = confidence interval; GR = triple-goal estimator; 2PL = two-parameter logistic model.

6.1. The Summary Is the Larger Lever. Which summary performed best was determined by the inferential goal, and the choice of summary moved the losses further

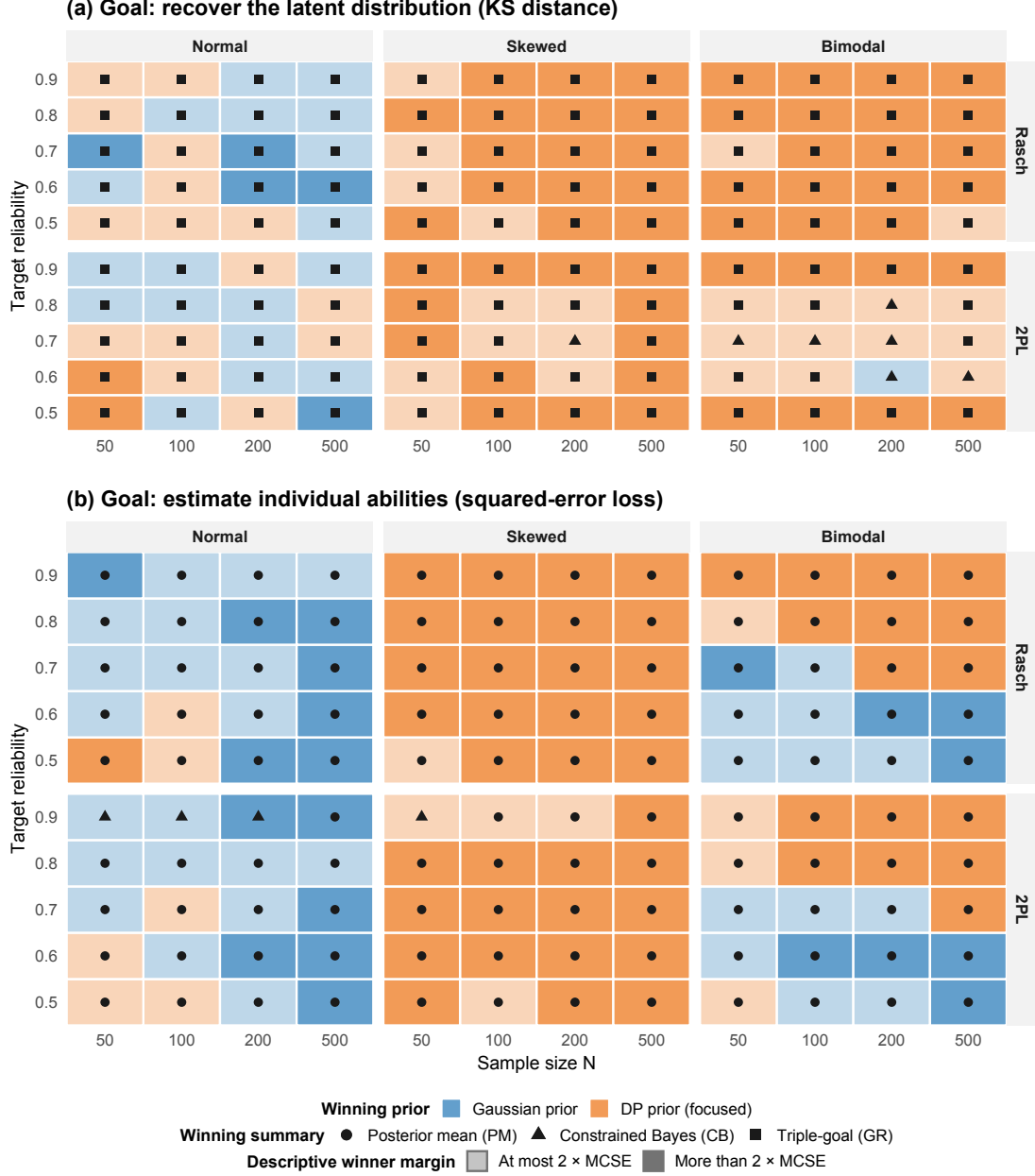
than the choice of prior. [Figure 2](#) shows, for each of the 120 conditions, which of the six prior-summary combinations won the distribution-recovery comparison and which won the individual-accuracy comparison. A GR combination won distribution recovery in 113 of 120 cells, and a PM combination won individual accuracy in 116 of 120, with the winning summary stable in about 97% of cells under exhaustive item-bank resampling; the pattern is thus close to categorical. Which prior colors the winning tile varies with the regime, but the summary column of the winner map is determined by the goal alone (censuses over the full nine-combination roster are given in [Appendix E](#)).

No single summary was therefore best for all purposes, and the registered H7 contrast quantifies the dissociation as a difference-in-differences: on non-normal cells, the PM-to-GR change in log loss on the registered distributional composite, minus the same change on individual squared error, is +0.281, a relative reversal of 32%, and every one of the 240 condition-by-prior points lies in the predicted quadrant. The two components are of unequal size: GR is 21% worse than PM on individual squared error, while PM is about 9% worse on the distributional composite, whose registered equal-weight construction includes the floor-bound tail family and therefore understates the distributional side of the reversal (with the tail family excluded, the descriptive gap is 0.66 log units; [Appendix D](#)). This is the empirical counterpart of the incompatibility result in [Section 3](#).

In [Figure 3](#) we compare, within each condition, how far the distributional losses move when the summary is swapped with the prior held fixed against how far they move when the prior is swapped with the summary held fixed. Swapping the summary moves KS loss by a median within-cell factor of 1.45, against roughly 1.1 for swapping the prior, and a factorial decomposition attributes 82% of the spread of cell-mean log KS losses to the summary main effect and 14% to the prior. The two mismatches are, moreover, not equally costly: using PM for a distributional goal costs 73% in KS loss at the two lowest reliability tiers (31% at the two highest), whereas using GR for an individual-accuracy goal costs 23% and 12% respectively. The mismatch that applied practice most often makes, reporting posterior means for a distributional purpose, is thus roughly three times as costly as the reverse one.

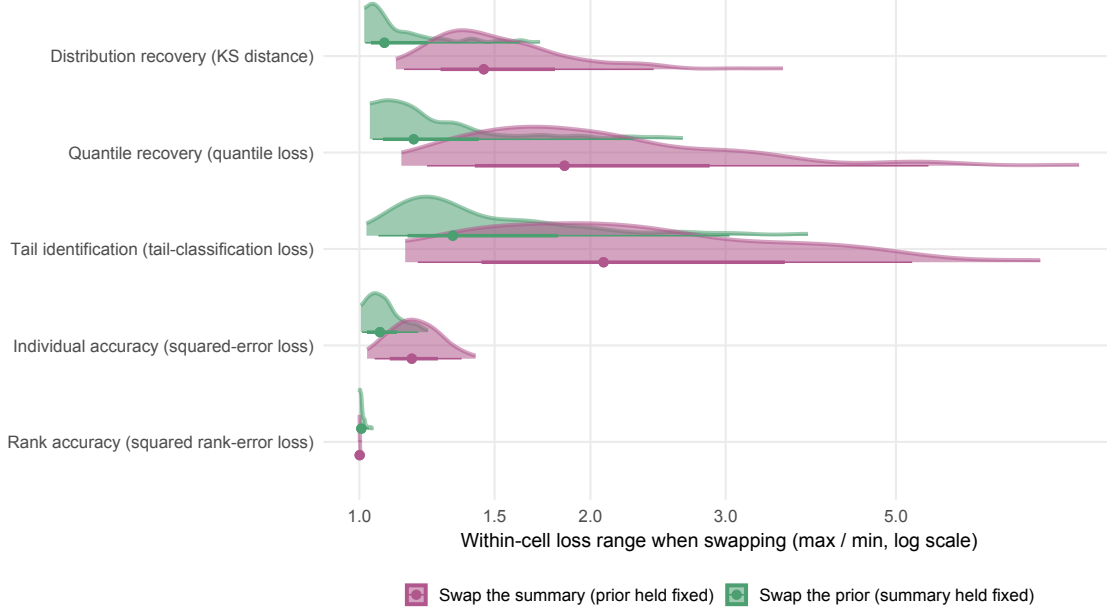
The two levers are not substitutes, however. Relative to the unimproved default (Gaussian prior with PM) on non-normal cells, in paired-geometric terms ([Section 5.4](#)), swapping the summary alone removes about 25% of KS loss, swapping the prior alone about 12%, and swapping both about 41%, more than the product of the

FIGURE 2. Winning Prior-Summary Combinations Across the 120 Conditions



Note. Each tile is one simulation condition (model family $\times$ latent shape $\times$ target reliability $\times$ sample size). Fill color gives the winning prior and the plotted symbol the winning summary among the six combinations of $\{\text{Gaussian, DP focused}\} \times \{\text{PM, CB, GR}\}$ for the goal named in each panel; lighter fill marks cells where the winner's margin over the runner-up is within twice its Monte Carlo standard error. The winner census is descriptive; confirmatory contrasts are in Table 3. DP = Dirichlet process; PM = posterior mean; CB = constrained Bayes; GR = triple-goal; KS = Kolmogorov–Smirnov distance; MCSE = Monte Carlo standard error.

FIGURE 3. Within-Condition Loss Ranges From Swapping the Summary Versus the Prior



Note. Distributions of the within-condition max/min loss ratio over the 120 conditions when one lever is varied and the other held fixed, by loss family. Points mark medians and the thick and thin bars the central 66% and 95% of conditions; the axis is log-scaled, and slab heights are normalized within each row for display. The tail-identification row is shown for completeness only; that loss family is floor-bound at this scale (Section 5.3) and supports no claims. Rank accuracy responds to neither lever. KS = Kolmogorov–Smirnov distance.

two single-lever gains (interaction -0.102 on the log scale). The crossed comparison shows the ordering directly: a Gaussian prior read through GR beats a focused DP prior read through PM in 100 of 120 cells under either ratio convention, and in 18 of the 20 exceptions the winner is DP with GR, so the exceptions reinforce rather than reverse the ordering. A flexible prior whose fit is read through a mismatched summary loses most of its benefit, because the summary discards the shape information the prior recovered before it reaches the report.

6.2. The Prior Follows the Data Regime. The flexible prior improved distribution recovery when the population was non-normal and the data were informative, and it was at near parity elsewhere. Figure 4 maps the primary contrast, the KS loss of DP(focused)+GR relative to Gaussian+GR, across all 120 cells. Averaged over non-normal cells, the flexible prior multiplies KS loss by .80 (H1); each doubling of N multiplies the ratio by a further .89 (H2); over the registered opportunity region

(non-normal, $N \geq 200$) the mean ratio is .73, and in the most favorable corner, where reliability .8 to .9 meets $N = 500$, the ratio reaches .48. The normal columns sit at parity apart from the three resolved cells already noted.

FIGURE 4. Distribution-Recovery Loss Ratios for the Primary Contrast

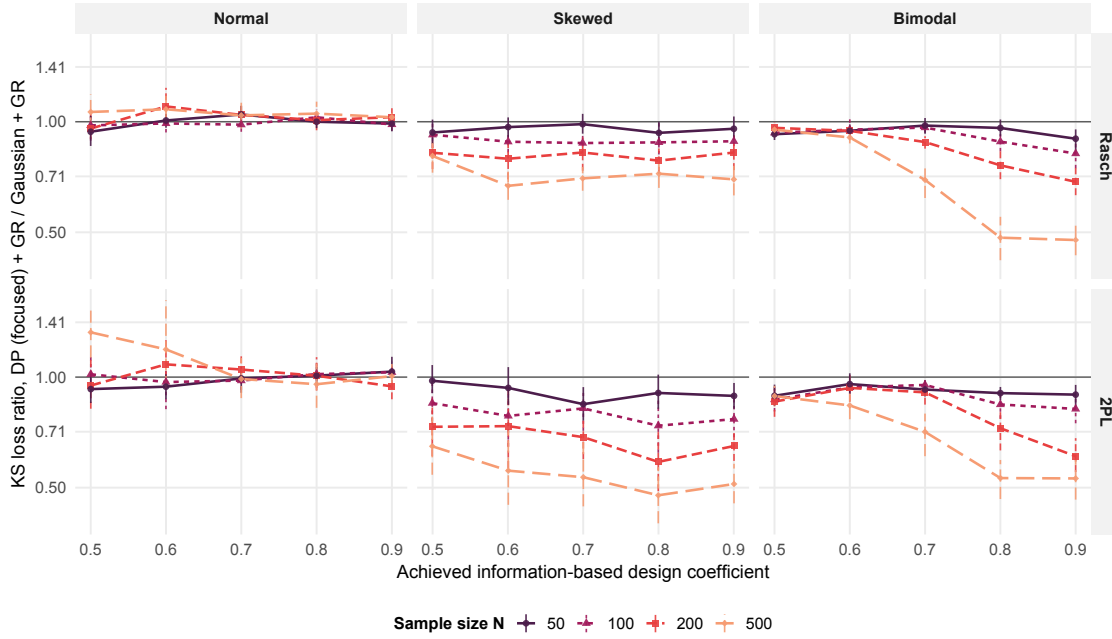


Note. Cell entries are ratios of mean KS loss, DP(focused)+GR relative to Gaussian+GR, over the 20 replicate data sets per condition; values below 1 favor the flexible prior. Outlined cells carry a registered win or strong-win evidence label; $\times$ marks the three normal-population cells whose interval resolves a real flexible-prior cost. DP = Dirichlet process; GR = triple-goal; KS = Kolmogorov–Smirnov distance.

The rate at which the advantage grows is governed by position along the realized ladder. The registered H6 slope of the log ratio on achieved reliability is -0.514 over non-normal cells, so a tenth of a point along the ladder, with the length and discrimination that accompany it, multiplies the advantage ratio by about .95; the same slope estimated within normal cells is -0.029 with an interval straddling zero, as would be expected of a genuine moderation on control data. Figure 5 traces the mechanism: for skewed populations the advantage is present at every tier and grows mainly with N , while for bimodal populations it fans out along the reliability axis, from near parity at $\bar{\rho} = .5$ to ratios near .48 at the top of the ladder, and

the exploratory shape split puts the bimodal slope at -0.775 against -0.253 for skew. Detecting two modes appears to require precise person estimates in a way that detecting a heavy right tail does not. Gains at the low tiers are not entirely absent, however: several bimodal cells at $\bar{\rho} = .5$ to $.6$ still show resolved gains of 9% to 16% at large N .

FIGURE 5. The Flexible-Prior Advantage as a Function of Achieved Reliability



Note. Mean KS loss ratio (DP focused + GR relative to Gaussian + GR) against achieved reliability, by sample size, shape, and model family; vertical bars are bootstrap 95% intervals. Ratios below 1 favor the flexible prior. The reliability axis is the achieved information-based design coefficient of Equation (3); because tiers were realized by a joint length-and-discrimination ladder, the slope reflects that bundle. DP = Dirichlet process; GR = triple-goal; KS = Kolmogorov-Smirnov distance.

6.3. Rasch Versus 2PL. The structure just described transfers across model families, with a registered qualification. The mean primary-contrast log ratio is -0.179 under Rasch and -0.270 under the 2PL, and the H3 interaction localizes the difference by shape: the Rasch advantage runs deeper in bimodal cells and the 2PL advantage in skewed cells (-0.127 , 95% CI $[-0.175, -0.078]$). A mechanism consistent with this pattern, which we offer as interpretation rather than as a tested claim, is that Rasch sum-score sufficiency transmits the shape of the score distribution directly into the

posterior, which helps a bimodal signal survive, while the 2PL’s estimated slopes can absorb part of an asymmetry before it reaches G . The families also differ in cost: at matched reliability the calibrated 2PL forms are about 10% shorter, but their samplers required three to five times the iterations of the Rasch fits. One boundary from [Section 2.4](#) travels with every 2PL result here: distribution recovery under the free-item 2PL is evaluated on the chosen scale representative, and nothing in it establishes convention-free identification of G .

6.4. The Individual-Accuracy Safety Screen. For individual scores the flexible pipeline was on average better, and in one region of the design systematically worse; both results are registered. On average, the registered non-inferiority test shows the flexible pipeline about 7% better on individual squared error, not merely non-inferior, over the opportunity region (H4, upper bound well inside the margin). The cell-level screen, however, carries a qualification that the average does not show, and [Figure 6](#) together with [Table 4](#) report it: 33 of the 40 opportunity-region cells pass the registered safety threshold, five carry caution labels, and two carry block labels. All seven flagged cells are bimodal populations at $\bar{\rho} = .5$ to $.6$ with $N \geq 200$, with individual-accuracy costs of 5% to 14% whose intervals exclude parity. The location of the flagged cells suggests the mechanism: the cost grows with N because the fitted G concentrates on the two modes as the sample grows, while the short response vectors remain too weak to allocate individual persons to a mode, so mid-scale respondents are pulled toward the wrong mode. Across all 120 cells, 48 improve on both goals under the flexible pipeline, 48 improve distribution recovery at some individual-accuracy cost, 23 worsen both (mostly normal cells, where nothing was there to gain), one sits on the quadrant boundary, and none improves individual accuracy while worsening distribution recovery.

The summaries carry their own individual-accuracy cost, which shrinks with information: at $\bar{\rho} = .5$ the median MSEL cost relative to PM is about 12% for CB and 20% for GR, falling to near zero and about 10% at $\bar{\rho} = .9$. When a distributional summary is reported for a low-reliability instrument, that cost should be stated next to the distributional gain, and [Section 8](#) does so.

6.5. Two Informative Null Results. Two comparisons produced essentially null results, and both are informative. First, rank accuracy is governed by the information regime alone: across conditions, the reliability tier (that is, position on the joint ladder) explains 94% of the variance in mean rank loss in a marginal regression, the

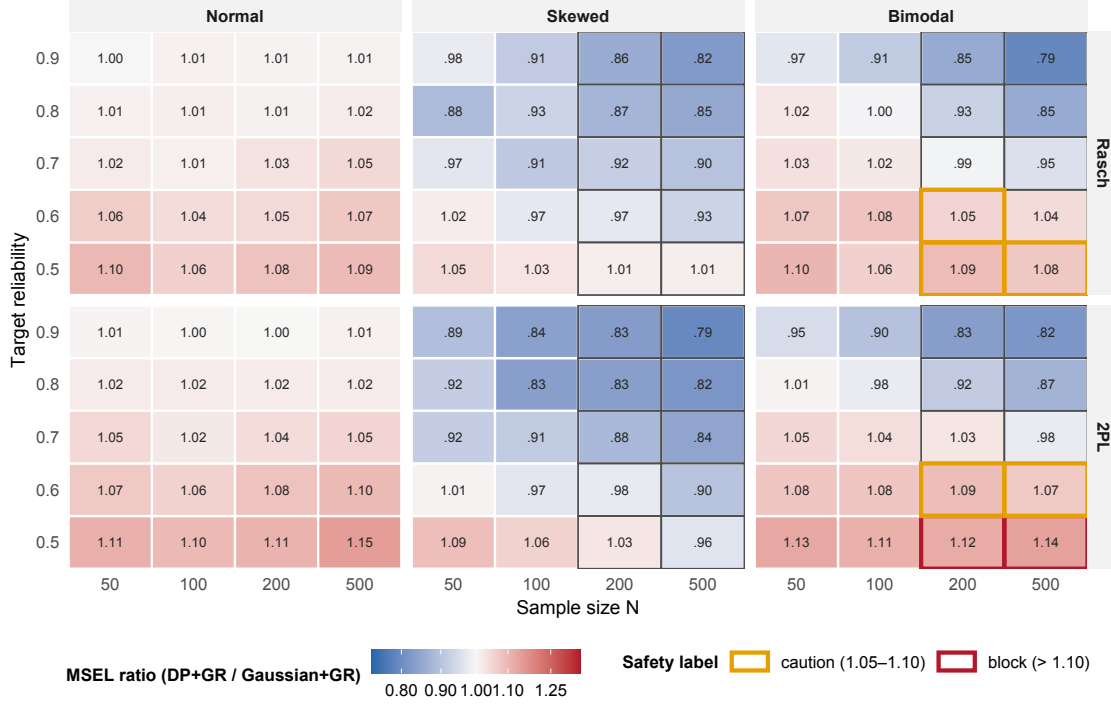
TABLE 4. The Seven Conditions in Which the Flexible Pipeline Costs Individual Accuracy

Condition (all bimodal)	N	$\bar{\rho}$	MSEL ratio	95% CI
2PL	500	.5	1.138	[1.115, 1.162]
2PL	200	.5	1.122	[1.095, 1.149]
Rasch	200	.5	1.094	[1.048, 1.136]
2PL	200	.6	1.092	[1.070, 1.116]
Rasch	500	.5	1.076	[1.059, 1.093]
2PL	500	.6	1.072	[1.059, 1.084]
Rasch	200	.6	1.052	[1.030, 1.077]

Note. MSEL ratio compares DP(focused)+GR with Gaussian+GR on individual squared-error loss; values above 1 favor the Gaussian pipeline. All seven cells are bimodal conditions at the two lowest reliability tiers with $N \geq 200$, and every interval excludes parity, so the pattern is systematic rather than imprecision. The disclosure rule requiring these cells to accompany any average no-harm statement was adopted after the results were seen and is flagged as such in the registration record ([Appendix D](#)). MSEL = mean squared error loss; DP = Dirichlet process; GR = triple-goal estimator; CI = confidence interval; 2PL = two-parameter logistic model.

identity of the six display-roster prior-summary combinations effectively none, and the largest within-condition spread across those combinations is a factor of 1.04. If ranking is the goal, the remedy is a more informative instrument, longer or better targeted, rather than a different estimator. Second, the two elicitation arms are nearly interchangeable in the posterior: the registered H5 contrast favors the focused arm by about 1%, the evidence map ties the arms in 106 of 120 cells, and under the strict convergence-pass sensitivity the estimate attenuates to -0.003 with an interval spanning zero. We read H5 as an indication that a defensible elicitation suffices rather than as strong evidence for preferring one arm. Convergence diagnostics were uniform across the 7,200 fits: all completed with no exclusions, the median of the per-fit worst $\hat{R}$ was 1.003, and effective sample sizes for the DP arms exceeded those of the Gaussian arms ([Appendix D](#)).

FIGURE 6. Individual-Accuracy Safety Screen Across the 120 Conditions



Note. Cell entries are ratios of mean individual squared-error loss (MSEL), DP(focused)+GR relative to Gaussian+GR; values above 1 indicate an individual-accuracy cost of the flexible pipeline. Outlined cells mark the registered caution (orange) and block (red) safety labels of Table 4; the framed region is the H4 opportunity region. DP = Dirichlet process; GR = triple-goal; MSEL = mean squared error loss.

7. The Findings on Real Tests

A simulation establishes which choice is better under known conditions, but not whether the choice makes a practical difference on the data sets analysts actually face. To examine the latter question we fitted thirteen real tests from the Item Response Warehouse under the paper’s full pipeline: each test under both the Rasch and 2PL models, each fit under all three priors, all three summaries read from every fit, at analysis samples of up to 500 persons drawn once with recorded seeds. The thirteen were chosen to span the simulation grid as far as real dichotomous data allow, from a national examination to short clinical scales; Table 5 lists them, and the screening rules and the placement of each test on the grid are given in Appendix F. Because the true abilities of real examinees are unknown, the case results are confined to statements of consequence: a choice differs, moves scores, reorders a top decile, or

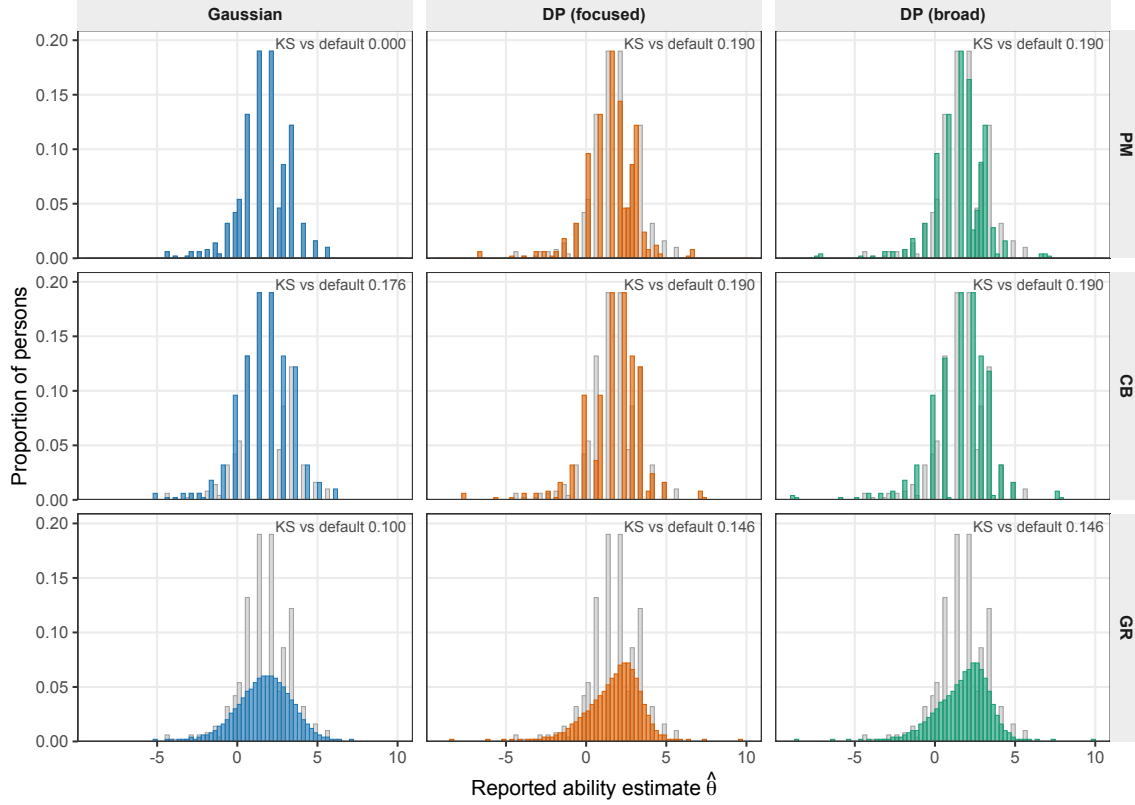
reclassifies respondents at a cutoff. Which direction is better is never inferred from the real data. Where a recommendation is wanted, it is looked up from the simulation at the test’s coordinates, its fitted reliability and its latent shape class, and that lookup is exploratory: it was assembled after both studies’ results were known. Rather than report all thirteen tests in turn, we present two cases in which the simulation’s findings appear in recognizable form and then summarize the portfolio.

7.1. A Bimodal Population Measured by a Short Test. The first case is a 16-item vocabulary checklist administered over the internet to 2,495 adults alongside a conspiracy-beliefs questionnaire (C4 in Table 5). Respondents either know a word or they do not, and the standardized screen places the fitted trait distribution in the bimodal class under the Rasch model at $\bar{\rho} = .77$: in the simulation’s terms, a bimodal population measured by a short, moderately reliable instrument.

Figure 7 shows the consequences of the two choices for this one data set. Each panel bins the reported scores of one prior-summary combination into common quarter-unit intervals, with the conventional default (Gaussian prior read through the posterior mean) repeated in every panel as grey bars. Reading down a column swaps the summary and changes the family of the report. The posterior means clump into eighteen narrow bands separated by empty stretches of scale, the largest band holding 95 of the 500 persons; CB pushes the same bands further apart; and GR replaces the bands with a smooth curve over the full range of the scale. Reading across a row swaps the prior and moves the scores themselves: the bands retain their membership but move along the scale, so the interval from 1.5 to 2.0, empty under the Gaussian prior, holds a 95-person band under the DP priors, and the reported ensemble sits at a KS distance of .19 from the default. These differences are not trivial in practical terms. Between-prior correlations of the scores exceed .996 here, yet on the portfolio’s bimodal cells the prior swap changes about 10% of a reported top decile, against a 6.2% floor that re-running the identical method with a new seed produces on its own; and because the report is banded, a band lying at a cutoff can move a cut share substantially while leaving every average nearly unchanged (at this test’s upper cutoff, by 19 percentage points).

Which of the nine should be reported is answered by the simulation’s matched cell (bimodal, $\bar{\rho} = .8$, $N = 500$, Rasch): a strong win for DP-with-GR on the distribution goal, at less than half the distributional loss of Gaussian-with-GR (ratio 0.48), while the posterior mean remains the winner for individual scores; against the matched cell’s best combination, the conventional default carries 2.54 times the distributional

FIGURE 7. Nine Reportable Distributions for One Real Test (C4, Rasch)



Note. Scaled empirical distributions (binned proportions, bin width 0.25) of the reported ability estimates for the vocabulary-checklist case (C4; 16 items, $n = 500$, bimodal placement, $\bar{\rho} = .77$) under each prior (columns) and summary (rows). The grey bars repeated in every panel are the conventional default report (Gaussian + PM), and the printed value is the KS distance between the panel's estimates and that default; no truth is available, so the comparisons describe consequences of the choice, not accuracy. PM and CB reports occupy narrow score bands with empty scale between them; GR reports spread over the full range. DP = Dirichlet process; PM = posterior mean; CB = constrained Bayes; GR = triple-goal; KS = Kolmogorov–Smirnov distance.

loss. Two cautions from the simulation also apply to this configuration. If the test had been less reliable, at tier .5 or .6 rather than .8, the flexible prior would have been expected to cost individual accuracy, since the flagged cells of Table 4 are bimodal populations on short low-reliability tests; and the 16-item Rasch form places C4 in the region where GR estimates can fail to reproduce themselves across seeds, so its GR report should be released only after a duplicate-seed agreement check (Section 7.3).

7.2. A Skewed Population at Operational Scale. The second case is the mathematics section of ENEM, Brazil's national secondary-school examination: nearly a

million examinees, of whom 5,000 were used to place the test on the grid and 500 to run the comparison (C7). The fitted trait distribution is strongly right-skewed under both models, the familiar shape of a hard test taken by a broad population, and this is the one high-stakes operational instrument in the portfolio.

For this case the consequences arise from the skewness of the fitted distribution rather than from a second mode. On the portfolio’s skewed cells the prior swap moves reported scores by a median of 0.136 standard deviations, the only movement in the portfolio above the prespecified 0.10 materiality bar, and the movement is concentrated where selection decisions operate: the lowest-scoring fifth of examinees moves up by 0.153 standard deviations on average while the highest-scoring fifth moves down, so the choice bears most heavily on the examinees whom a cutoff or an intervention screen would affect. At the matched simulation cells the conventional report carries 2.41 times the matched winner’s distributional loss under the Rasch fit and 2.40 under the 2PL, the largest factors in the portfolio, and both are strong-win cells for the flexible pipeline. ENEM also illustrates a measurement issue that arises before any prior-summary choice: moving from the Rasch to the 2PL model changes this test’s EAP-based marginal reliability from .96 to .78 while the design coefficient [Equation \(3\)](#) moves from .66 to .84, in the opposite direction. Which number is “the reliability of the test” depends on the functional, a disagreement we return to below.

7.3. Across the Thirteen Tests. Across the portfolio the transported verdicts reproduce the structure of the simulation results. [Table 5](#) collects the thirteen tests, each fitted under both models, with the winning combination for the distribution goal transported from each case-cell’s matched simulation condition. Of the 26 test-by-model cells, 19 carry a determinate shape class inside the grid’s support and receive a verdict (median distance between fitted reliability and the matched tier .033). The winners follow the simulation’s pattern: a GR combination wins the distribution goal in 18 of 19 cells, all 14 non-normal cells inherit strong wins for the flexible pipeline, with distribution-recovery ratios of 0.48 to 0.76 at their matched cells, and the five normal-control cells land at parity, ratios of .99 to 1.04, which is the outcome expected of a sound lookup on control cells. For the other two goals a table is unnecessary: a posterior-mean combination wins individual scores in all 19 cells, and on the ranking goal essentially every cell is a near-tie, reproducing the simulation’s rank result in the real data. Against the prespecified Gaussian-with-GR comparator the conventional default carries a median of 1.20 times the distribution-recovery loss on these cells; against the winner selected at each matched cell, an optimistic post-hoc benchmark,

the median is 2.08. The matched cells, loss ratios, and evidence labels behind each verdict are given in [Appendix F](#).

Two further findings emerged from the real data that the simulation could not have produced. The first concerns reliability itself. Across the thirteen tests, moving from the Rasch to the 2PL model changes EAP-based marginal reliability by a median of +0.215 but changes the design coefficient [Equation \(3\)](#) by only +0.004, and the two functionals move in the same direction in only 6 of 13 tests; the shape class, too, flips between models for 6 of the 13. A statement like “this test has reliability .8 and a bimodal population” is, at this scale of data, partly a statement about the fitted model, and any recommendation keyed to such coordinates should be checked under both models. The second finding concerns reproducibility. GR person estimates failed a seed-to-seed self-reproduction check, median movement above a tenth of a reference standard deviation, on 11 of 78 fits, every failure a Rasch fit of a 12-to-16-item form, while the posterior mean and CB reproduced within 0.03 standard deviations everywhere; ten of the eleven failures had unremarkable convergence diagnostics, so the instability is attributable to the summary rather than to the sampler. On short Rasch forms, GR should not be reported from a single run. The mechanism, the full consequence analysis, and the reproducibility record are given in [Appendix F](#).

TABLE 5. Thirteen Real Tests From the Item Response Warehouse and the Winning Combinations for the Distribution Goal

Case	Instrument	N_{tot}	n	I	Rasch		2PL	
					$\bar{\rho}$ (class)	Winner	$\bar{\rho}$ (class)	Winner
C1	Content-literacy reading (Grade 3, US)	1,914	500	29	.84 (sk)	DP+GR	.86 (sk)	DP+GR
C2	Vocabulary, literacy intervention	2,118	500	12	.76 (sk)	DP+GR	.73 (sk)	DP+GR
C3	Inductive reasoning test (Brazil)	1,803	500	56	.95 (bi)	DP+GR	.93 (sk)	DP+GR
C4	Vocabulary checklist (internet sample)	2,495	500	16	.77 (bi)	DP+GR	.26 (sk) ^a	—
C5	PIRLS reading extract (Grade 4)	3,480	500	35	.87 (no)	G+GR ^b	.88 (no)	G+GR ^b
C6	NAEP extract (US)	1,510	500	12	.70 (no)	G+GR ^b	.71 (no)	DP+GR ^b
C7	ENEM 2013 mathematics (Brazil)	999,822	500	45	.66 (sk)	DP+GR	.84 (sk)	DP+GR
C8	Millon Clinical Multiaxial Inventory extract	1,208	500	44	.88 (bi)	DP+GR	.87 (sk)	DP+GR
C9	Fraction subtraction test	536	500	20	.90 (nn)	—	.73 (bi)	DP+GR
C10	Verbal aggression questionnaire	316	316	24	.85 (bi)	DP+GR	.85 (bi)	DP+CB
C11	Story recall task	235	235	32	.89 (un)	—	.89 (un)	—
C12	ZAREKI number processing (clinical, children)	341	341	16	.42 (bi) ^a	—	.56 (un)	—
C13	Work Design Questionnaire	1,055	497	18	.65 (no)	G+GR ^b	.65 (un)	—

Note. N_{tot} = respondents available in the warehouse; n = analysis sample (drawn once with a recorded seed; capped at 500 by the elicitation routine); I = items. $\bar{\rho}$ is the information-based design coefficient computed from each fitted model, and the shape class comes from the standardized screen with case-specific null calibration: sk = skewed, bi = bimodal, no = normal-consistent, nn = non-normal without a determinate class, un = undetermined. Six of the 13 cases change shape class between the two models. Winner is the combination with the smallest mean distribution-recovery loss among the six prior-summary combinations at the case-cell’s matched simulation condition, transported by the exploratory lookup of [Section 7](#); dashes mark the seven cells that either lack a determinate shape class or fall below the grid’s support. A posterior-mean combination wins the individual-score goal in all 19 eligible cells, and the ranking goal is a near-tie everywhere. The matched cells, loss ratios, and evidence labels are given in [Table F.1](#). ^aReliability below the simulation grid’s support; these two cells receive no transported recommendation. ^bNormal-control cells: the matched-cell ratios are .99–1.04, so the verdict is parity within Monte Carlo error and the printed combination is not a recommendation. DP = Dirichlet process (focused); G = Gaussian; PM = posterior mean; CB = constrained Bayes; GR = triple-goal; 2PL = two-parameter logistic model; IRW = Item Response Warehouse (Domingue et al., 2025).

8. Practical Guidance and Software

The results reduce to a short decision sequence, applied in order. The first question is what the report is for. If the deliverable is individual scores, the posterior mean should be reported, and the prior matters little except in the flagged cells discussed below. If the deliverable is a ranking or a selection, no choice of estimator will improve it; the effective remedy is a more informative instrument, longer or better targeted. If the deliverable describes the group (a distribution, a spread, a share below a cutoff), a distribution-matched summary is the single most consequential choice: GR where its conditions hold, CB when only a scale correction is wanted or when GR's conditions fail. Two caveats accompany any GR recommendation. On Rasch forms of about 16 items or fewer, GR should not be reported from a single run; a duplicate-seed agreement check, or CB in its place, is the remedy ([Section 7.3](#)). And GR's individual-accuracy cost, about 20% at reliability .5 and about 10% at .9, should be stated alongside the distributional gain when both kinds of number will be read from the same report.

The second question is where the instrument lies on the reliability scale. The tier thresholds that follow are positions on our realized ladder, on which reliability, test length, and discrimination move together, so an instrument should be placed by its own $\bar{\rho}$ with that bundle in mind. At tiers up to about .7, the Gaussian prior with a goal-matched summary is a defensible default: most of the achievable improvement comes from the summary, the flexible prior's distributional gain is modest except for skewed populations at large N , and the seven flagged bimodal cells (MSEL ratios 1.052–1.138, every interval excluding parity; [Table 4](#)) counsel against accepting an individual-accuracy cost for it. From tier .8 upward, when non-normality is plausible on substantive grounds, the focused DP prior combined with GR was the best observed distributional pipeline in our grid, with loss ratios reaching one half, and it improved individual accuracy at the same time except in that flagged region. Where the population is credibly normal, the flexible prior offers no gain and the Gaussian prior carries no cost. When the shape class itself drives a decision, both models should be fitted, since six of our thirteen real cases changed class between Rasch and 2PL; disagreement between the two fits should be read as information about model sensitivity rather than as noise.

These recommendations are implemented in three open-source R packages that form one pipeline. `IRTsimrel` generates reliability-targeted item response data for

design work (Lee, 2026f); DPprior elicits the concentration hyperprior from cluster-count beliefs (Lee, 2026c); and DPMirt fits the Rasch and 2PL models under Gaussian or DP mixture priors in NIMBLE and returns all three summaries from one fit (Lee, 2026b). A minimal analysis reads

```
library(DPMirt)
alpha <- dpmirt_alpha_prior(N = 500, mu_K = 25, confidence = "medium")
fit    <- dpmirt(resp, model = "rasch", prior = "dpm",
                 alpha_prior = alpha, nchains = 4,
                 niter = 12000, nburnin = 4000, seed = 1)
est    <- dpmirt_estimates(fit, methods = c("pm", "cb", "gr"), seed = 1)
```

after which `est$theta` holds the PM, CB, and GR ensembles for reporting and `dpmirt_diagnostics(fit)` the convergence and clustering summaries. A complete worked example on one of the case-study data sets, from elicitation through fitting to the duplicate-seed GR check, is given in [Appendix G](#). The main practical obstacle is computational cost: in the case studies a DP fit took 15 to 17 times as long as its Gaussian counterpart (medians of roughly 230 against 14 minutes at $n \leq 500$ under our conservative four-chain budgets), and the elicitation routine’s exact-moment machinery is designed for analysis samples of at most 500 persons, which is why the case studies score at most 500 examinees per fit.

9. Discussion

We have evaluated, jointly and against three inferential goals, the two levers available to an analyst reporting person parameters from a Bayesian item response model: the prior on the trait distribution and the posterior summary read from the fit. A simulation over 120 reliability-targeted conditions in the Rasch and 2PL models, run under an analysis plan locked and hash-fingerprinted before the production fits, joined by thirteen real data sets from the Item Response Warehouse analyzed under the same pipeline, supports three general conclusions: the inferential goal determines the appropriate summary; the data regime determines the appropriate prior; and the instrument’s information governs how much either choice matters.

The role of test reliability. Position on the reliability ladder governed every result in this study: the flexible prior’s advantage steepens along it, the summaries’ mismatch penalties shrink along it, and rank accuracy answers to nothing else. Because the ladder was realized by jointly varying test length and discrimination, these are

effects of that bundle, which is also what a reliability difference is in practice: instruments differ in $\bar{\rho}$ because they differ in length, targeting, and discrimination. Three practices follow. Test designers deciding where a marginal item or a better-targeted item pool goes should treat distributional reporting goals as an argument for information investments, since the achievable gains from any estimator compound with the instrument’s information. Analysts should estimate and report reliability before choosing a pipeline, naming the coefficient: the case studies showed the two common functionals disagreeing by a fifth of their range on the same fits, and recommendations keyed to “reliability .7” are not interpretable until the estimand is stated. And simulation studies of estimation methods should treat the reliability of the generated data as an explicit design quantity, stating how it was realized; the reliability-targeted machinery used here makes that routine.

Two levers, one estimator. The estimator that reaches a reader is a prior-summary pair, and our results counsel treating the summary as the first choice rather than as an afterthought. Reading a distributional report off posterior means costs 31% to 73% in distribution-recovery loss depending on the tier; reading individual scores off GR costs 12% to 23%. The asymmetry implies that the common practice of defaulting to the posterior mean for every purpose is also the costly direction of error. At the same time the levers compound: on non-normal populations the best pipeline pairs the flexible prior with the matched summary, and a flexible prior read through the wrong summary loses most of its distributional benefit. The useful unit of advice is therefore a combination of model, summary, and instrument rather than a model alone. The same division of labor was reported by Lee et al. (2025) for multisite trials; what this study adds is the demonstration that it survives transport into item response models, where the posteriors the summaries read are built from dichotomous responses with jointly estimated, convention-identified item parameters, under an information level the instrument’s designer controls.

When flexible priors help, and when they cost. Over most of the design the Dirichlet process mixture behaved as intended for a safeguard specification: near-parity on normal populations at a certified average cost of 1.6%, distributional gains of up to half the loss where non-normal populations met informative data, and on average a 7% individual-accuracy improvement in the registered region. The one systematic failure is informative: in bimodal populations measured at reliability .5 to .6 with hundreds of examinees, the fitted prior concentrates on the two modes more quickly than the short response vectors can assign persons to them, and individual

accuracy deteriorates by 5% to 14% (the seven cells of [Table 4](#), every interval excluding parity). Applied users facing that configuration, a plausibly clustered population measured by a short instrument, should keep the Gaussian prior for individual scores even if they adopt the flexible pipeline for the distributional report.

Limitations. Five boundaries on these conclusions deserve emphasis. First, the reliability factor is a realized ladder of jointly varying test length and discrimination, with the two nearly collinear by construction; effects of $\bar{\rho}$ are effects of that bundle, and a design that varies discrimination at fixed length remains to be run. Second, the simulation’s populations are three stylized shapes, its item banks one balanced template per family, and its scope dichotomous unidimensional models without guessing, missingness, or person misfit; the recommendations are hypotheses about transport to other designs rather than general properties of the estimators. Third, we studied point summaries only; interval estimates, coverage, and decisions computed from full posteriors are untouched. Fourth, the tail-share loss family was uninformative at this scale by construction, so no claims about extreme-tail recovery are supported. Fifth, the join of case coordinates to simulation cells was assembled after both studies’ results were known, and the case portfolio was drawn from a warehouse that over-represents shareable, curated, largely low-stakes data.

Extensions. The clearest next step is polytomous response models, which the elicitation and summary machinery already accommodates conceptually but the fitting software does not; about three quarters of the shape-measured warehouse corpus is polytomous, so this is where the practical demand lies. Multidimensional and diagnostic classification models pose the same goal-summary questions in richer parameter spaces, and the identification analysis of [Section 2.4](#) suggests that semiparametric versions of those models will need their constraint conventions settled before their fitted distributions can be read. Within the present scope, the open theoretical question we consider sharpest is the identification of G under a 2PL with jointly estimated items, for which no counterpart of the Rasch-case results is available.

The choices studied in this paper are made, implicitly or explicitly, whenever person parameters are reported from an item response model. Making them explicit requires little: the analyst should name the goal, match the summary to it, place the instrument on the reliability scale, and reserve the flexible prior for settings informative enough to support it.

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Software and reproducibility. All Bayesian model fits used NIMBLE (de Valpine et al., 2017) through the DPMirt package (Lee, 2026b); concentration priors were elicited with DPprior (Lee, 2026c); and the simulated item-response data were generated with IRTsimrel (Lee, 2026f) under R 4.6.0 (R Core Team, 2026). The three packages are available from GitHub (<https://github.com/joonho112/DPMirt>; <https://github.com/joonho112/DPprior>; <https://github.com/joonho112/IRTsimrel>), each with a documentation site under <https://joonho112.github.io/>. A replication package at <https://github.com/joonho112/dpmirt-replication> rebuilds every number, figure, and table in this paper and its supplement from the frozen analysis data, and carries the random seeds, sampler budgets, convergence records, the locked analysis plan, and annotated reference code for the production fits, which are documented rather than shipped. Three companion open-access monographs document the study in more detail than an article allows: the simulation study (<https://joonho112.github.io/dpmirt-simulation-study/>), the theoretical development (<https://joonho112.github.io/dpmirt-theory-book/>), and the case studies (<https://joonho112.github.io/dpmirt-case-studies/>).

Data availability. All item-response data analyzed in the case studies are public through the Item Response Warehouse (IRW; Domingue et al., 2025), <https://datapages.github.io/irw/>. Original response-level data remain governed by the licenses of the primary IRW sources and are not redistributed here.

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Online Supplement

Targeting Toward Inferential Goals in Bayesian Item Response Models for Estimating Person-Specific Latent Traits

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This supplement accompanies the main text. Section, equation, table, and figure numbers below are prefixed by the appendix letter (A.1, B.1, ...). References to the main text use its unprefix numbering.

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Appendix A. Model and Computation Details

A.1. The Hierarchical Model as Fitted. All fits in the simulation and the case studies used the following specification, which we state exactly as implemented in `DPMirt` because several of its choices differ from textbook defaults in ways that matter for interpretation. At level 1, responses follow [Equation \(1\)](#). At level 2, the traits follow [Equation \(4\)](#) with either the Gaussian or the DP mixture prior on G . At level 3, the item difficulties carry a centered prior: latent difficulties are drawn as $\tilde{\beta}_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$ with $\sigma_\beta^2 = 3$ fixed, and the fitted difficulties are $\beta_i = \tilde{\beta}_i - \bar{\tilde{\beta}}$, which enforces the identification constraint $\sum_i \beta_i = 0$ inside the sampler. The induced prior on $(\beta_1, \dots, \beta_I)$ is therefore singular on the sum-to-zero hyperplane, with $\text{Var}(\beta_i) = \sigma_\beta^2(1 - 1/I)$ and pairwise correlation $-1/(I - 1)$; there is no mean parameter μ_β and no hyperprior on σ_β^2 . Under the 2PL the discriminations receive a log-normal prior centered as described in [Appendix B](#). At level 4, the Gaussian arm’s hyperpriors are $\mu_\theta \sim N(0, 3)$ and $\sigma_\theta^2 \sim \text{InvGamma}(2.01, 1.01)$, a specification with prior mean 1.0 and large but finite variance rather than the pathological $\text{InvGamma}(\varepsilon, \varepsilon)$ form criticized by Gelman ([2006](#)).

For the DP mixture arms, the base measure on kernel means and variances is $H_0 = N(0, \sigma_\mu^2) \times \text{InvGamma}(\nu_1, \nu_2)$ with $(\sigma_\mu^2, \nu_1, \nu_2) = (2, 2.01, 1.01)$, the `DPMirt` defaults, adapting the computational template of Paganin et al. ([2023](#)), who used $\sigma_\mu^2 = 3$ with the same inverse-gamma parameters. The mixture is sampled in its

marginal partition representation (Section 4.1) with truncation level $M = 50$, raised to 75 at $N = 500$ in the focused arm and up to 150 at the larger sample sizes in the broad arm and the case studies; the truncation guard was reached in 5 of 4,800 DP fits in the simulation, and those fits carry the diagnostic on record. H_0 is a prior on cluster locations, not on θ : its variance of 2 does not assert $\text{Var}(\theta) = 2$, and centering it does not center G (Appendix B).

A.2. Sampler Budgets and Convergence. Table A.1 lists the Markov chain Monte Carlo budgets, which were fixed in advance of production; the 2PL budgets were raised once, before any production fit, after a pilot mixing failure, under a registered raise-only rule. All fits ran four chains.

TABLE A.1. Sampler Budgets per Model-Prior Configuration

Configuration	Iterations	Burn-in	Thin	Retained draws
Rasch, Gaussian	8,000	4,000	2	8,000
Rasch, DP	12,000	4,000	2	16,000
2PL, Gaussian	40,000	20,000	10	8,000
2PL, DP	36,000	12,000	6	16,000

Note. Retained draws pool four chains. DP = Dirichlet process (both elicitation arms); 2PL = two-parameter logistic model.

Convergence screening excluded nothing. All 7,200 simulation fits completed; none met the hard failure criteria; 7,014 passed the strict screen and 186 (2.6%) carried soft warnings, concentrated in the 2PL arms (the Rasch DP arms produced none). The median across fits of the per-fit worst $\hat{R}$ was 1.003. Median bulk effective sample sizes were ordered with the DP arms (about 5,100 to 6,800) above the Gaussian arms (about 3,300 to 5,600), reflecting the DP arms’ larger retained pools. Posterior cluster occupancy in the focused arm grew with sample size, as expected for a parsimony prior: pooled median occupied clusters of 3, 4, 7, and 14 at $N = 50, 100, 200$, and 500. In the case studies, 4 of 78 fits crossed $\hat{R} = 1.1$ on the monitored parameters (worst 1.28); they were retained with their diagnostics on record, and the seed-replication analysis of Section 7.3 covers them.

A.3. Computation. The simulation’s fitting run comprised 7,200 fits in 147.7 host-hours across a 12-worker laptop and a 44-worker cloud allocation, with a fit archive of 174 GB; the operating-characteristic analysis added under three hours. In the case studies a DP fit took 15 to 17 times as long as its Gaussian counterpart (median 230.0 against 13.2 minutes under Rasch, 222.4 against 14.9 under the 2PL), and the

production plus seed-replication runs totaled 364 core-hours. These figures reflect deliberately conservative budgets; practitioners fitting a single data set at $n \leq 500$ should expect run times of hours rather than days on current hardware.

Appendix B. Identification Details

B.1. Orbits and Constraints in the Parametric Models. Identification statements below follow San Martín (2016). Under the Rasch model the observationally equivalent transformations form a one-dimensional location orbit, $(\theta_p + c, \beta_i + c)$ for any constant c ; the unit discrimination leaves no scale freedom. Under the 2PL the orbit is two-dimensional: for any $a > 0$ and c ,

$$\theta_p^* = a(\theta_p - c), \quad \beta_i^* = a(\beta_i - c), \quad \lambda_i^* = \lambda_i/a \quad (\text{B.1})$$

reproduces every response probability, so a location and a scale restriction are both required, and the scale restriction is what pins down the scale of G itself.

Which restrictions are needed depends on the specification. With fixed (incidental) person parameters, $\beta_1 = 0$ suffices under Rasch, and $\beta_1 = 0$ with $\lambda_1 = 1$ under the 2PL. With random effects and a G of known shape up to location and scale, the Rasch model needs no scale restriction at all: σ_θ is identified from the data through the strictly increasing map from (σ_θ, β) to the bivariate response probabilities. Setting $\sigma_\theta^2 = 1$ in that family therefore restricts an identified parameter, which is a modeling assumption capable of being wrong, not the removal of an indeterminacy. Location constraints, by contrast, are genuine conventions, and they act on different objects in different specifications: in the marginal random-effects model the choice is between anchoring an item ($\beta_1 = 0$), centering the items ($\sum_i \beta_i = 0$), or fixing a location functional of G ($\mu_\theta = 0$). A constraint on realized person draws such as $\sum_p \theta_p = 0$ is not a constraint on the marginal parameter. When G is an estimand, item centering is the informative choice because it leaves the location of G free and estimable; fixing $\mu_\theta = 0$ instead forces every fitted G to be centered by construction and hides one of the quantities of interest. We also note that a proper posterior does not imply identification: proper priors produce proper posteriors along unidentified orbits, and credible intervals along such orbits reflect the prior.

B.2. The Semiparametric Case. Two results of San Martín et al. (2011) govern the DP mixture setting. First, the location indeterminacy persists and is shared between the item parameters and the location of the infinite-dimensional G ; centering the base measure H_0 fixes only the prior expectation of G , and a realized draw from the process

has a nonzero mean almost surely, so using “the prior is centered” as an identification device biases the item parameters. DPMirt accordingly refuses ability-side constraints under a DPM prior and defaults to item centering, which is exact ($\mathbf{1}c = 0$ implies $c = 0$) and leaves G entirely free. Second, identification of G itself is partial on any finite test: with one item parameter anchored, the identified functionals of G under the Rasch model are the $I + 1$ integrals

$$\xi_G(k) = \int \frac{e^{k\theta}}{(1 + e^\theta) \prod_{j \geq 2} (1 + e^{\theta - \beta_j})} dG(\theta), \quad k = 0, \dots, I, \quad (\text{B.2})$$

a test-tilted Laplace transform of G at integer points; full identification requires $I \rightarrow \infty$. A posterior over G from a 25-item test therefore constrains 26 functionals, and the prior supplies the rest, which is why the elicitation of [Appendix C](#) deserves design-level care in a way it would not in a parametric model.

B.3. The 2PL Under a DPM Prior, and Two Software Regimes. For the 2PL with jointly estimated items and a DPM prior on G , no identification theorem comparable to the Rasch results is available; a finite known-item 2PL identifies at most the 2^I response-pattern probabilities, which do not point-identify G over unrestricted measures, and the sharp characterization with free items is, as far as we are aware, an open problem. We therefore treat the 2PL DPM’s fitted distribution as convention-dependent and report sensitivity rather than claiming identified recovery.

Operationally, DPMirt offers two ways to impose the 2PL conventions, and the two studies used one each, which we record because they are not the same Bayesian model even though they select the same likelihood representative. The in-model regime (`constrained_item`) centers the difficulties and centers the log discriminations during sampling, so that $\sum_i \beta_i = 0$ and the geometric mean of λ_i is 1 at every iteration; the production simulation used this regime. The post-hoc regime (`unconstrained` with `rescale = TRUE`) samples without constraints and then applies, to every retained draw s , the transformation [Equation \(B.1\)](#) with $c_s = \bar{\beta}_s$ and a_s the geometric mean of the sampled discriminations, so that the transformed discriminations λ_i/a_s have unit geometric mean and every draw lands on the same representative; the case studies used this default. In the first regime the priors act on the constrained space, in the second on the unconstrained one, so posterior geometry can differ even though the reported representative is common; the within-study comparisons in both studies are unaffected because every arm within a study shares its study’s regime.

Appendix C. Concentration-Prior Elicitation

C.1. The Concentration Parameter and the Cluster Count. Conditional on α , the number of occupied clusters K_P among P persons follows the distribution of Antoniak (1974),

$$\mathbb{P}(K_P = k \mid \alpha) = |s(P, k)| \frac{\alpha^k \Gamma(\alpha)}{\Gamma(\alpha + P)}, \quad k = 1, \dots, P, \quad (\text{C.1})$$

where $|s(P, k)|$ are unsigned Stirling numbers of the first kind, with conditional mean $\mathbb{E}(K_P \mid \alpha) = \alpha\{\psi(\alpha + P) - \psi(\alpha)\} \approx \alpha \log\{(P + \alpha)/\alpha\}$ for the digamma function ψ . The expected cluster count grows logarithmically in P , and K_P is sufficient for α , so beliefs about how many distinct trait clusters a design of size P should reveal translate directly into information about α . At the same time, matching cluster-count beliefs alone does not control the mixture weights: under $\alpha \sim \text{Gamma}(a, b)$ the first stick-breaking weight satisfies, in closed form,

$$\mathbb{P}(w_1 > t \mid a, b) = \left\{ \frac{b}{b - \log(1 - t)} \right\}^a, \quad (\text{C.2})$$

and a hyperprior tuned only to $\mathbb{E}(K_P)$ can still place substantial mass on a single dominant component, the unintended-prior phenomenon described by Lee (2026a); Vicentini and Jermyn (2025) raise the same concern through the prior distribution of the stick-breaking weights. (In the size-biased stick-breaking order, w_1 is the asymptotic mass of the cluster containing a randomly chosen person, not necessarily the largest component, so its tail probabilities bound dominance from below.)

C.2. The Two-Stage Moment-Matching Procedure. The design-conditional framework of Lee (2026a) takes as input the design size P , a target mean μ_K for K_P , and a target variance σ_K^2 (or a confidence label mapped to one), and returns the $\text{Gamma}(a, b)$ hyperparameters reproducing those moments exactly. Stage 1 uses the shifted-Poisson proxy $K_P - 1 \mid \alpha \approx \text{Poisson}(\alpha \log P)$, under which the Gamma mixture is negative binomial and moment matching is closed-form: with $\mu_0 = \mu_K - 1$,

$$a_0 = \frac{\mu_0^2}{\sigma_K^2 - \mu_0}, \quad b_0 = \frac{\mu_0 \log P}{\sigma_K^2 - \mu_0}. \quad (\text{C.3})$$

Stage 2 refines (a_0, b_0) by Newton iteration in $(\log a, \log b)$ against the exact finite- P prior-predictive moments of Equation (C.1), computed by Gauss–Laguerre quadrature; the refined solution matches the elicited moments under the true induced distribution, not the proxy. The weight diagnostics Equation (C.2) are reported as part of

the output, and a dual-anchor variant re-solves for (a, b) under a penalized compromise when a stated bound on $\mathbb{P}(w_1 > t)$ conflicts with the cluster-count targets. The framework’s simulations document why the care is warranted: a vague $\text{Gamma}(1, 1)$ hyperprior collapses the posterior to one or two clusters in about 60% of replications across true cluster structures, and abundant data do not rescue it.

C.3. The Two Arms as Configured. Both studies carried two elicitation arms defined by a common rule mapping the analysis sample size N to a cluster-count target, $\mu_K(N) = \min\{c_{\max}, \max(c_{\min}, sN)\}$, with the variance set by the confidence label ($\sigma_K^2 = \text{VIF} \times (\mu_K - 1)$), where the label medium maps to $\text{VIF} = 2.5$ and low to 5.0). The focused arm, the primary candidate throughout the paper, used $(c_{\min}, s, c_{\max}) = (4, 0.05, 25)$ with medium confidence, so its targets were $\mu_K = 4, 5, 10$, and 25 at $N = 50, 100, 200$, and 500; the broad arm used $(6, 0.1, 50)$ with low confidence, doubling the slope, the cap, and the variance inflation, so its targets were 6, 10, 20, and 50 across the same ladder. [Table C.1](#) gives the resulting hyperparameters and weight diagnostics at each design size (the elicitation is performed at $P = N$); the values match the frozen production configuration, and any row is reproduced by a single `DPprior` call, as illustrated in [Appendix G](#). The diagnostics also record a limitation at the ladder’s lower end: at $N = 50$ the cluster-count targets sit near their feasibility floor and both arms place substantial prior mass on a dominant component ($\mathbb{P}(w_1 > .5) = .60$ and $.48$), a property the H5 comparison shows the posterior tolerates. The released package tabulates the exact Antoniak distribution with Stirling-number tables up to a design cap of 500 persons and its input validator enforces that cap for every fit, which is the origin of the case studies’ analysis-sample cap; the cap is a software restriction, not a mathematical one, since the Stage-2 moment equations are defined for any P through digamma and trigamma functions, and the Stage-1 closed form depends on P only through $\log P$.

TABLE C.1. The Two Elicitation Arms as Configured, With Induced Weight Diagnostics

Arm	N	μ_K	σ_K^2	a	b	$\mathbb{P}(w_1 > .5)$	$\mathbb{P}(w_1 > .9)$
Focused	50	4	7.5	1.148	1.243	.601	.300
Focused	100	5	10.0	1.666	1.646	.557	.233
Focused	200	10	22.5	3.529	1.648	.290	.046
Focused	500	25	60.0	8.904	1.626	.042	.0004
Broad	50	6	25.0	0.610	0.304	.484	.270
Broad	100	10	45.0	1.179	0.400	.306	.105
Broad	200	20	95.0	2.454	0.428	.094	.011
Broad	500	50	245.0	6.220	0.443	.003	.0000

Note. μ_K and σ_K^2 are the elicited mean and variance of the occupied cluster count K_N at the design size; (a, b) is the Gamma hyperprior on the concentration parameter returned by two-stage moment matching, which reproduces the targets exactly. $\mathbb{P}(w_1 > t)$ is the induced tail of the first stick-breaking weight, the size-biased mass of the cluster containing a randomly chosen person.

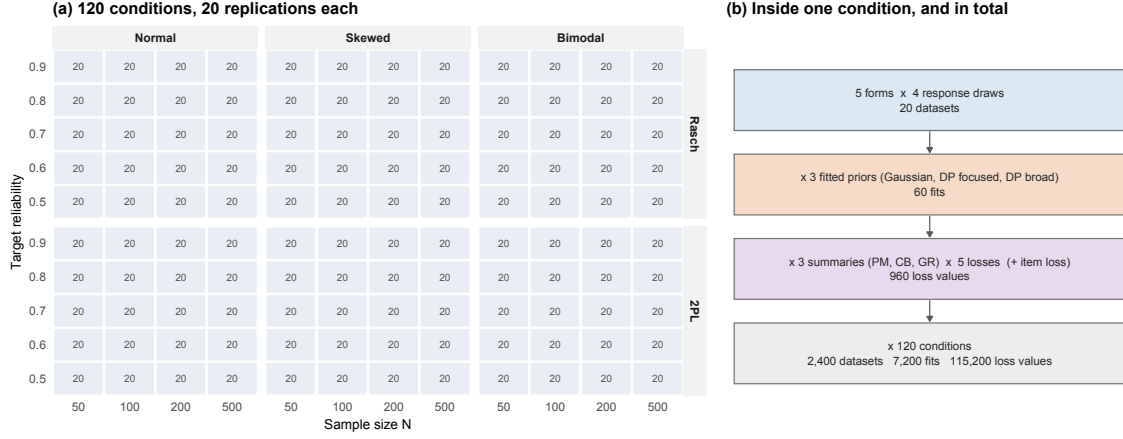
Values computed with DPprior and identical to the frozen production configuration.

Two properties of this configuration were confirmed by the simulation rather than assumed. The focused arm’s parsimony did not constrain the posterior’s cluster count (occupied clusters grew from 3 to 14 across the N ladder; [Appendix A](#)), and the choice between the two arms mattered by about 1% on the primary loss, with even that margin specification-dependent ([Section 6.5](#)). A defensible elicitation therefore suffices; finer tuning of the hyperprior had no measurable benefit in this design.

Appendix D. Simulation Details

D.1. Design Realization. [Figure D.1](#) summarizes the design’s multiplication chain: 120 conditions, each with five item banks and four response draws (2,400 data sets), three prior arms per data set (7,200 fits), and sixteen loss records per fit (115,200 rows). The reliability ladder was realized by strictly nested item subsets per model and bank, refined by the cell-specific global discrimination multiplier c^* ([Section 5.2](#)); across the 150 calibration cells the frozen c^* values span 0.794 to 1.090, the achieved design coefficient matches its target within .003 everywhere, and the relative range of c^* across shapes within a tier has median 1.9% and maximum 5.1%. Because the ladder couples length and discrimination, the centered achieved reliability and the centered log item count correlate .98 across cells; a companion regression that enters both (variance inflation factor 31) yields -0.670 (SE 0.228) on reliability and $+0.021$ (SE 0.030) on log items, which we report as a description of the realized bundle and explicitly not as a causal partition.

FIGURE D.1. The Design Grid and Its Multiplication Chain



Note. Rows and columns display the crossed factors; the chain beside the grid tracks conditions to data sets to fits to loss records. 2PL = two-parameter logistic model.

D.2. Registration Record. The analysis plan was locked on 2026-07-24, before any production fit, and its SHA-256 fingerprint accompanies every confirmatory row. Six addenda followed. Addenda 001 and 002 (reference grid for H1; symmetric floors for replicate-level log ratios) predate the pilot’s completion; Addendum 003 with its amendment registered the secondary H6–H7 family after the pilot but before any production fit; Addendum 004 (an optional replication extension) was written mid-wave, blind to outcomes, and its trigger did not fire (5 of 40 evaluable cells ambiguous, below the .2 threshold); Addendum 005 (a common reporting-scale repair and the restriction of H6 to the realized ladder) followed the fits but preceded the production analysis. Addendum 006 requires separate mention: it was adopted after the production outcomes were known, when the individual-accuracy safety screen fired, and it constrains reporting only, requiring the seven flagged cells to be disclosed, with ratios and intervals, wherever the H4 average is stated. It changed no threshold and no test, and [Table 4](#) in the main text implements it.

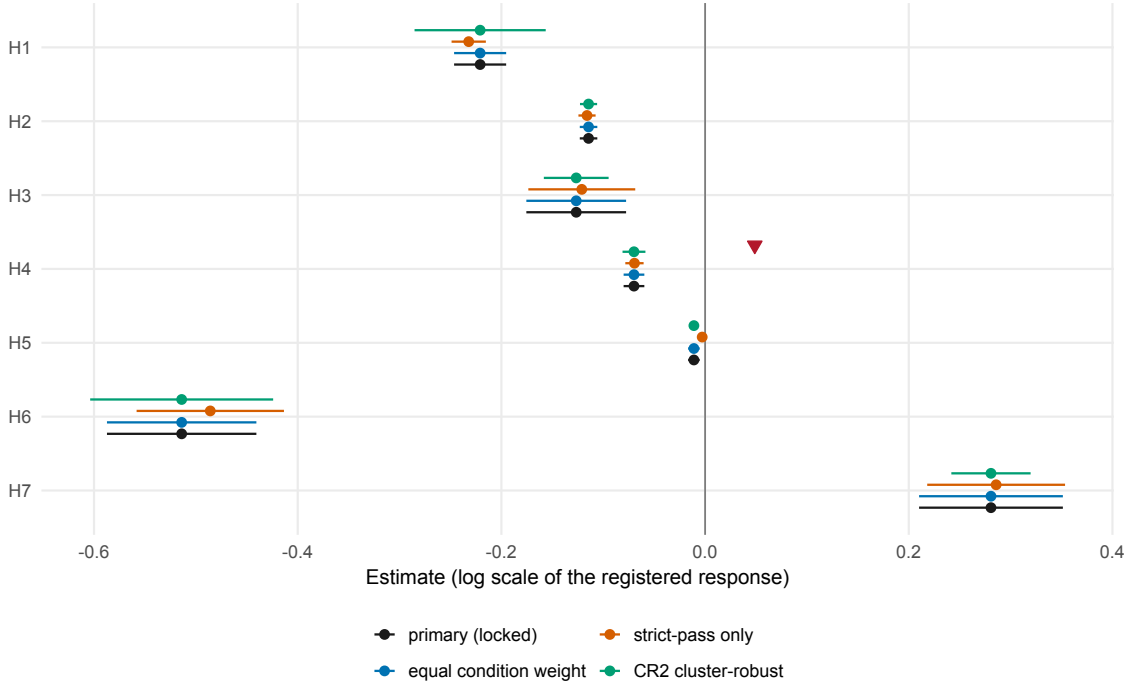
D.3. Confirmatory Machinery and Operating Characteristics. Two ratio conventions appear in the paper’s reporting, and every use names its convention. Cell-level ratios attached to the registered evidence labels ([Figures 4 and 6](#), [Table 4](#), and the transported verdicts of [Section 7](#)) are ratios of equal-form condition means with hierarchical bootstrap intervals; the compounding and crossed-lever summaries of [Section 6.1](#) are paired-geometric ratios, exponentiated means of within-replicate log

ratios. The two agree closely (the largest log-scale disagreement across the 120 primary cells is about .04, and four of 240 primary and crossed cells change side of parity under the alternate convention), but they are different estimands, and no headline count or rounded effect depends on the choice.

Each registered contrast is a mean of replicate-paired log loss ratios (or a slope of such ratios), estimated by a linear mixed model with fixed effects for shape, $\log_2 N$, centered achieved reliability, and model family with registered interactions, and a random intercept for item-bank family; the Holm correction runs over exactly the five primary p -values, with the secondary pair corrected separately. Per-cell evidence labels come from a hierarchical bootstrap with $B = 10,000$: a cell is a win when its ratio is at most 0.95 with the interval below 1 (a strong win when the ratio is at most 0.90 with upper bound at most 0.95 and an absolute improvement of at least .01), a tie when point and interval lie in $[0.95, 1.05]$, and a flag otherwise, the flag being a residual category rather than a verdict. Replication adequacy met its target in all six registered contrasts (median interval half-widths .020 to .078 against the .100 target). The realized operating-characteristic analysis, run after the wave with 400 simulations per grid point, places type-I rates at .035 to .048 against the nominal .05, the normal-cell false-win rate at .008 against the .05 cap, and power at 1.0 for every hypothesis by true ratio 0.85 (all observed effects sit at 0.80 or beyond).

D.4. Sensitivity. Figure D.2 displays every registered estimate under four specifications: the primary weighting, equal condition weights, a strict convergence-pass subset, and cluster-robust (CR2) standard errors. H1 through H4, H6, and H7 move by a few percent of their values across specifications. The one exception is H5, whose estimate attenuates from -0.011 to -0.003 under the strict-pass subset with a two-sided interval spanning zero; we accordingly describe the focused-versus-broad margin as reassurance about elicitation insensitivity rather than as a finding with independent standing. One further sensitivity concerns H7: its registered distributional composite weights the three distributional families equally and therefore includes the floor-bound tail family, whose component enters with the opposite sign; recomputed descriptively over the KS and quantile families alone, the reversal contrast is 0.66 log units rather than 0.28, so the registered estimate is the conservative one and the reversal claim does not lean on the tail family.

FIGURE D.2. Registered Estimates Under Four Analysis Specifications



Note. Points are estimates and horizontal bars 95% intervals for H1–H7 under each specification; the vertical line marks zero effect, and the red triangle on the H4 row marks the registered non-inferiority margin ($\log 1.05$), which the H4 interval clears in every specification. The H4 average is qualified by the seven flagged cells of Table 4, whose intervals exclude parity. CR2 = cluster-robust standard errors; the CR2 rows for the secondary family are diagnostic only.

Appendix E. Additional Simulation Results

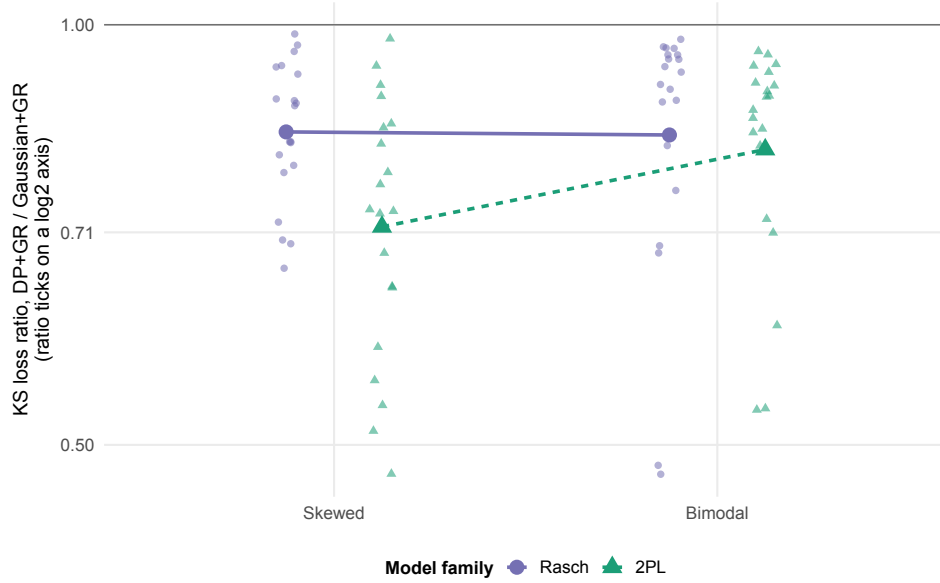
E.1. Winner Censuses Over All Nine Combinations. The winner maps of Figure 2 use the six-combination roster (two priors by three summaries). Over the full nine-combination roster, including the broad arm, the distribution-recovery census is DP(focused)+GR in 47 of 120 cells, DP(broad)+GR in 44, Gaussian+GR in 23, DP(broad)+CB in 5, and DP(focused)+CB in 1, the near-even split between the two DP arms reflecting that they tie in 106 of 120 cells; the individual-accuracy census is DP(focused)+PM in 51, Gaussian+PM in 50, DP(broad)+PM in 15, Gaussian+CB in 3, and DP(focused)+CB in 1. The summary identity of the winner is the stable element (96.7% and 98.8% of cell-resamples under exhaustive item-bank resampling of the six-combination display maps); the prior tile is the regime-dependent one (about 94% stable), and we treat the prior tiles as provisional in the sense that a different draw of item banks could move individual cells without moving the census.

E.2. Mismatch Penalties and the Cost of Each Summary. Read as a penalty matrix over the two reliability bands, using PM for a distributional goal costs 73% in KS loss over the two lowest tiers and 31% over the two highest, and 133% and 60% respectively on the quantile family; using GR for an individual-accuracy goal costs 23% and 12%. Rank penalties are near zero for every summary. The tail family’s nominal winner shifts erratically across tiers, which is the label churning expected of a floored loss ([Appendix D](#)) and one reason we make no tail claims. Relative to PM within the same fit, the summaries’ own individual-accuracy cost falls with information: the median MSEL increase is about 12% (CB) and 20% (GR) at reliability .5, and reaches the tie band (CB) and about 10% (GR) at .9.

E.3. The Crossed-Lever Comparison and Its Exceptions. Gaussian+GR beats DP(focused)+PM in 100 of 120 cells under either ratio convention. The 20 exceptions are concentrated where information is abundant and the population is not normal, and in 18 of them the overall winner is DP+GR (DP+CB in the other two), so the exceptions mark cells in which the prior’s contribution has become large enough to add to the summary’s, rather than cells in which the summary ordering reverses. The two levers’ gains compound: on non-normal cells, summary swap alone -25% KS, prior swap alone -12% , both -41% , against -34% if the effects were multiplicative (log interaction -0.102).

E.4. Shape Geography in Detail. The exploratory shape split behind [Figure 5](#) estimates the ladder slope at -0.775 (SE 0.053) in bimodal cells against -0.253 (SE 0.052) in skewed cells. For skewed populations the advantage appears early: it is present at reliability .5 for every N , reaching a ratio near .65 at the 2PL with $N = 500$ even on the lowest tier. For bimodal populations it appears later but is not absent at low tiers: resolved gains include 2PL ratios of .857 (95% CI [.780, .937]) at reliability .5 with $N = 200$ and .837 ([.768, .902]) at reliability .6 with $N = 500$, and a Rasch ratio of .906 at reliability .6 with $N = 500$. [Figure E.1](#) displays the registered H3 interaction: the Rasch advantage runs deeper in bimodal cells and the 2PL advantage in skewed cells, the 2PL profile rising toward the flat Rasch profile without intersecting it.

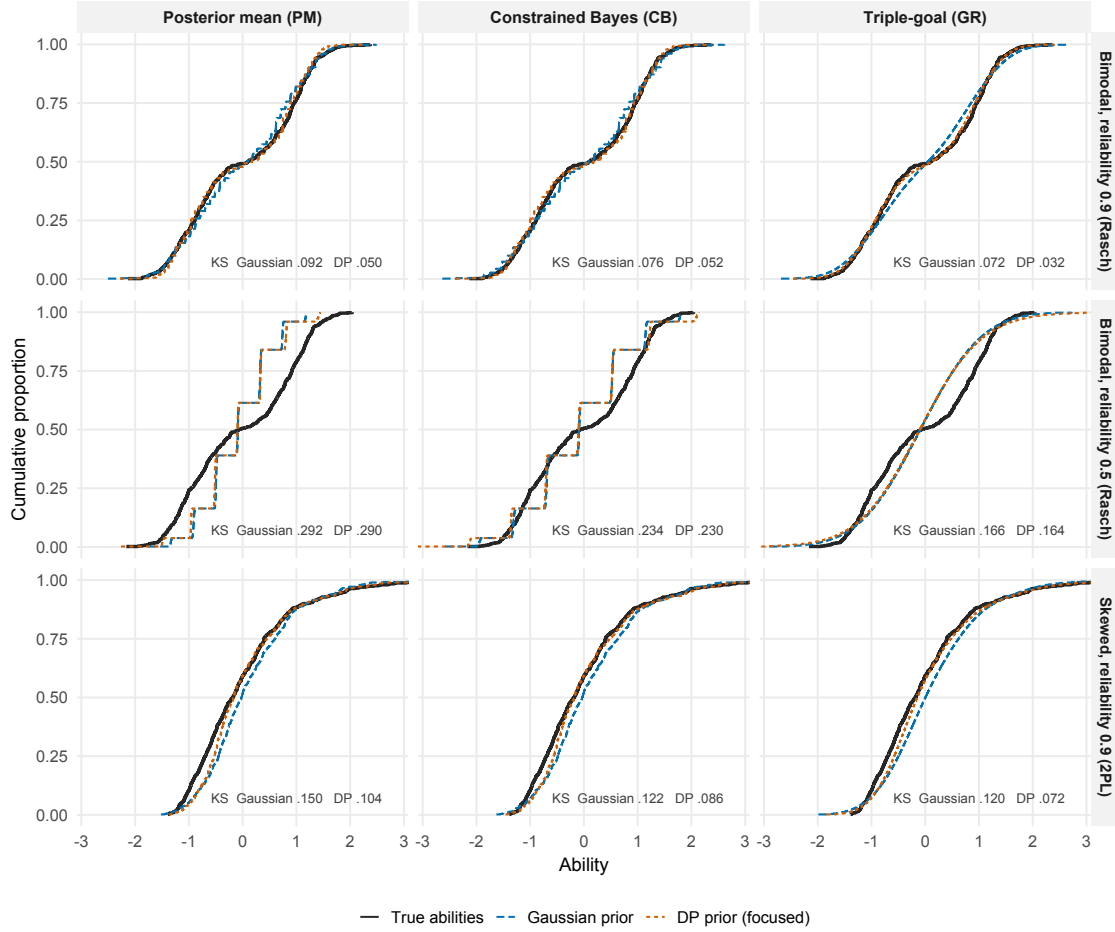
FIGURE E.1. The Model-Family by Shape Dissociation (H3)



Note. Mean primary-contrast log KS ratios by model family and latent shape; the diverging slopes (near-flat under Rasch, rising under the 2PL) are the registered H3 interaction (-0.127 , 95% CI $[-0.175, -0.078]$). KS = Kolmogorov–Smirnov distance; 2PL = two-parameter logistic model.

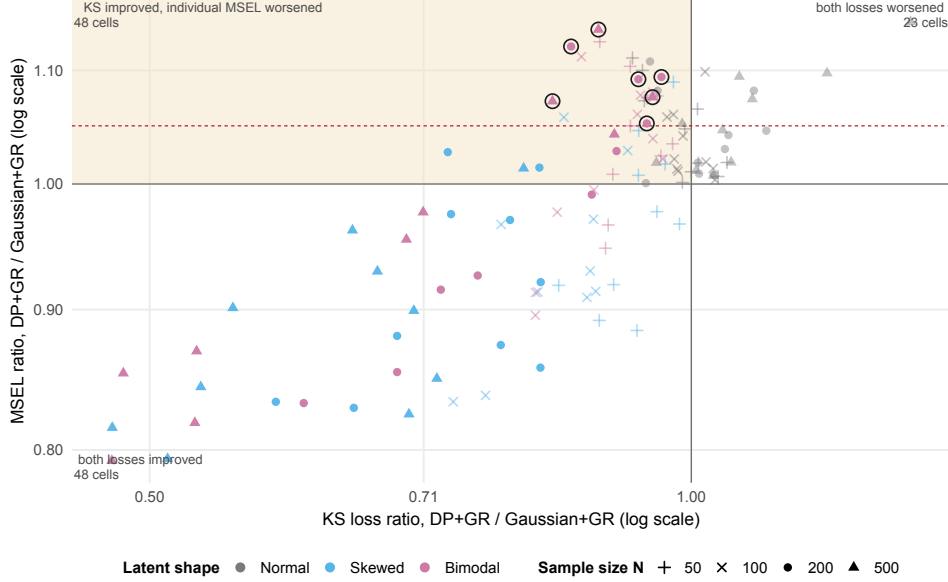
E.5. Mechanism Exhibits. Figure E.2 shows estimate-set EDFs against the realized truth for three exemplar cells. In the favorable bimodal corner (reliability .9, $N = 500$) the flexible pipeline’s EDF tracks the two modes with a KS distance of .032 against .072 for Gaussian+GR on the same replication; at reliability .5 under Rasch, the PM and CB ensembles collapse into score-associated bands while GR restores a full-range curve. Figure E.3 gives the goal-tradeoff census across all 120 cells: 48 cells improve on both distribution recovery and individual accuracy under the flexible pipeline, 48 improve the first at some cost to the second, 23 worsen both (predominantly normal cells), none improves individual accuracy while worsening distribution recovery, and one sits on the boundary.

FIGURE E.2. Estimate-Set EDFs Against Realized Truth in Three Exemplar Cells



Note. Colored curves are the EDFs of the six prior-summary combinations' reported estimate sets; the black curve is the EDF of the realized true abilities for that replication, and the printed values are KS distances to it. The three panels are single exemplar replications (bimodal at reliability .9 and $N = 500$ under both families; bimodal at reliability .5 under Rasch), chosen to display the mechanism, not to represent their cells. EDF = empirical distribution function; KS = Kolmogorov-Smirnov distance.

FIGURE E.3. Distribution-Recovery Gains Against Individual-Accuracy Costs



Note. Each point is one condition, plotting the flexible pipeline’s log MSEL ratio (vertical) against its log KS ratio (horizontal); the dashed line marks the 1.05 safety boundary and the circled points are the seven flagged opportunity-region cells. Quadrant counts are given in the text. MSEL = mean squared error loss; KS = Kolmogorov–Smirnov distance.

E.6. Rank Flatness and the Quantile Family. Across the 120 conditions, a marginal regression of mean rank loss on reliability tier attains $R^2 = .940$, while the corresponding regression on the identity of the six display-roster combinations attains R^2 below 10^{-6} , and the largest within-condition max/min rank-loss ratio across those combinations is 1.039; these are descriptive decompositions, not variance partitions, but their disparity is the rank-flatness result. The quantile family reproduces the KS geography in amplified form (ratios reaching .25; GR wins 96 of 120 cells), with the caveat that four normal-control cells show extreme ratios (up to 2.22) produced by near-zero denominators; the prose therefore quotes KS.

Appendix F. Case-Study Details

F.1. Corpus, Screen, and Provenance. The portfolio funnel ran 996 catalogued warehouse units, to 889 with a reliability measurement in the companion census (Lee, 2026e), to 572 that completed the latent-shape measurement pipeline (504 of which enter the published diagnostic frame of Lee (2026d)), to 138 dichotomous units, to 29

survivors of the eligibility rules: a single administration, at least 1,000 respondents, 12 to 60 items, approximate unidimensionality, and a converged flexible calibration. From these the 13-test portfolio (Table 5) was assembled to span the simulation grid rather than to represent any population: of 60 screenable units, 28 sit in the .75–.85 reliability band and only 6 below .65, so the two lowest-reliability cases were recruited deliberately, three small data sets were fitted whole, and one sub-threshold benchmark (C9, the fraction-subtraction data set of 536 respondents) was admitted for its grid coverage and scored at 500. Each test was fitted at an analysis sample of at most 500 persons drawn once with a recorded seed (the elicitation routine’s design cap), with a separate characterization sample of up to 5,000 used only to place the test on the grid: 78 production fits, each replicated at a second seed.

The provenance of the cases required attention: the warehouse’s catalogue fields were wrong or empty for three of the thirteen cases, which were identified from their source publications, and roughly a third of the corpus carries no usable license field, so each case dossier records a verbatim dataset citation and the catalogue status at freeze. The corpus over-represents shareable, curated, largely low-stakes data; the one high-stakes operational exam in the portfolio (C7, ENEM mathematics) was characterized on a 5,000-person subsample of its million examinees.

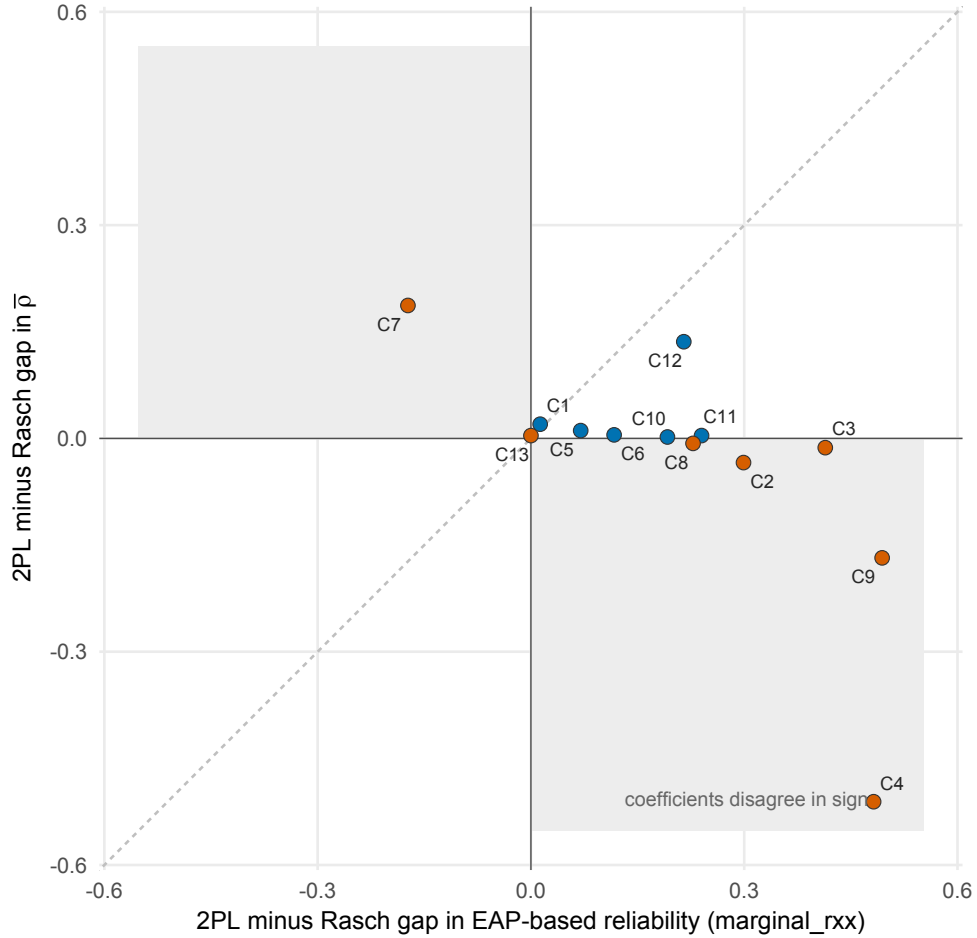
Materiality thresholds for the four consequence families were frozen before any fit: individual scores are material when the median absolute movement exceeds 0.10 reference SD; the reported distribution when the between-report KS distance exceeds .05 or the SD ratio leaves $[0.95, 1.05]$; rankings when more than 5% of ranks move by more than two percentile points or the top-decile Jaccard overlap falls below .90; and cut shares when any fixed cut’s share moves by more than two percentage points. Scales were never standardized before comparison. One limitation of the thresholds is recorded rather than repaired: the cut-share rule was aimed at net movement and does not register reclassification that cancels, as the C4 band-edge episode showed.

F.2. The Shape Screen and Its Standardized Statistic. Fixed-threshold shape screens are unreliable at these sample sizes: under exact normality, a fixed multimodality threshold flagged 98% of $n = 100$ replicates in our null calibration, and 47 of 54 small warehouse data sets under the 2PL. The screen therefore calibrates every statistic against the case’s own null (200 refits of data simulated from a normal population with the case’s n , test length, and item parameters) and additionally requires the observed departure to reach at least half the simulation condition’s, so a class label indicates a departure large enough for the simulation grid to be informative,

not a statistically significant departure from normality. The multimodality statistic itself had to be rebuilt once: the original implementation computed the dip on each fitted model’s native scale with a fixed absolute smoothing bandwidth, while fitted density SDs ranged from 0.51 to 2.15 across the 26 case-cells, making observed values, nulls, and the simulation’s reference condition mutually incommensurable. The standardized refit (26 observed and 5,158 null replicate refits under identical settings) left all scale-invariant statistics unchanged, reproduced 25 of 26 frozen labels, and moved one borderline cell (C9 under Rasch, undetermined to non-normal without class), a change the legacy estimator reproduces as well, so the correction mattered for commensurability and not for conclusions.

[Figure F.1](#) documents the reliability-coordinate finding stated in the main text: across the 13 cases the Rasch-to-2PL change in EAP-based marginal reliability (median +0.215) and in the design coefficient of [Equation \(3\)](#) (median +0.004) agree in sign in only 6 cases, and one case (C4) moves by half the axis in opposite directions under the two functionals.

FIGURE F.1. Disagreement Between Two Reliability Functionals on the Same Fits



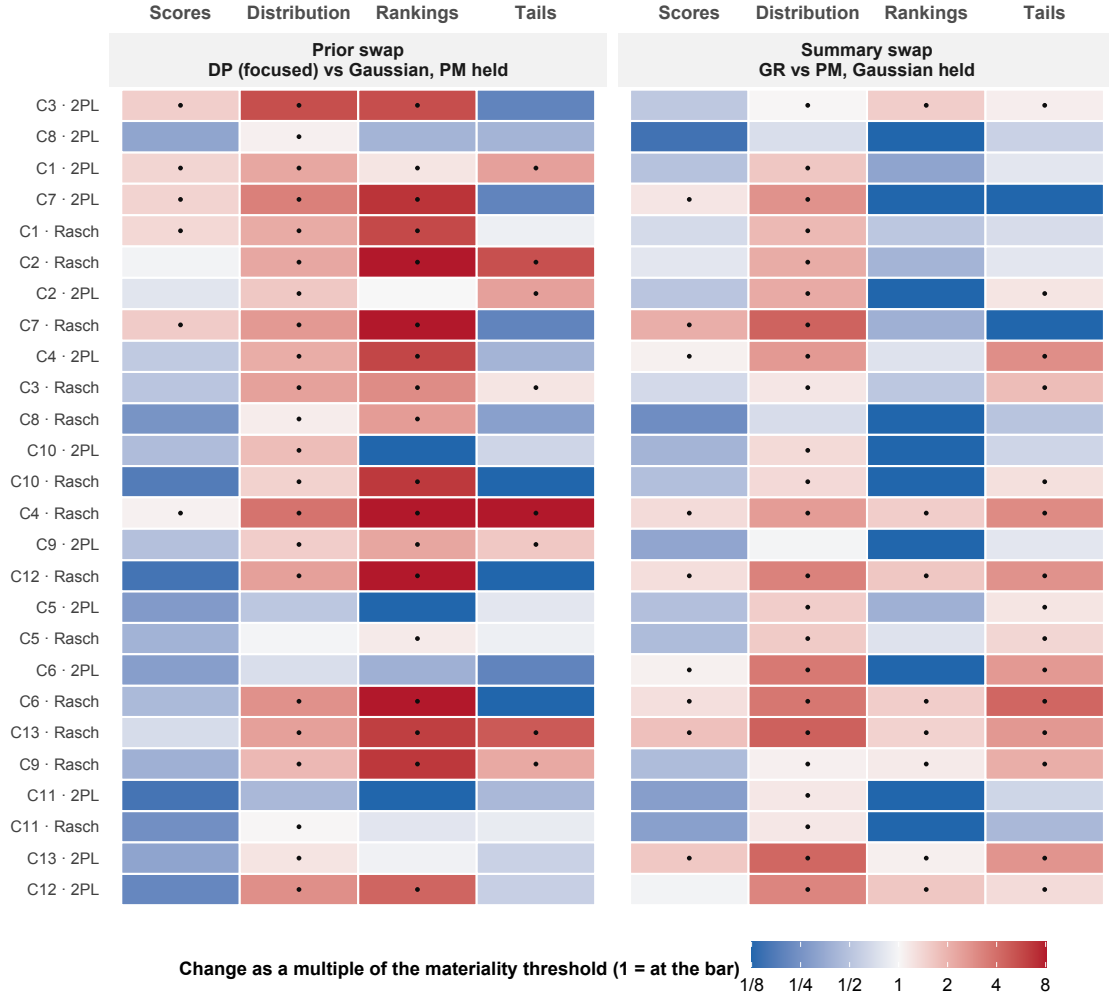
Note. Each point is one case, plotting its Rasch-to-2PL change in EAP-based marginal reliability (horizontal) against the same change in the information-based design coefficient $\bar{p}$ (vertical); the dashed line is agreement, and points in the shaded off-diagonal quadrants (orange) are the seven cases whose two functionals disagree in sign. C4 moves by half the axis in opposite directions.

EAP = expected a posteriori.

F.3. Consequence Results in Detail. Figure F.2 organizes all 26 case-cells by the four consequence families and the two levers, colored by multiples of the materiality threshold. The score-versus-distribution inversion is visible at a glance: bimodal cells are material on the distribution in seven of seven but on scores in one of seven, skewed cells on both (scores median 0.136 SD; distribution median KS .106), and normal controls mostly on neither. On the distribution family the change is a change of family rather than of scale: on skewed cells the DP ensemble is wider in SD (ratio

1.06) yet narrower in interquartile range (0.93), with excess kurtosis moving from -0.17 to $+1.14$; after within-ensemble standardization the priors' EDFs still separate on non-normal cases and coincide on the controls.

FIGURE F.2. Materiality of the Two Levers Across All 26 Case-Cells

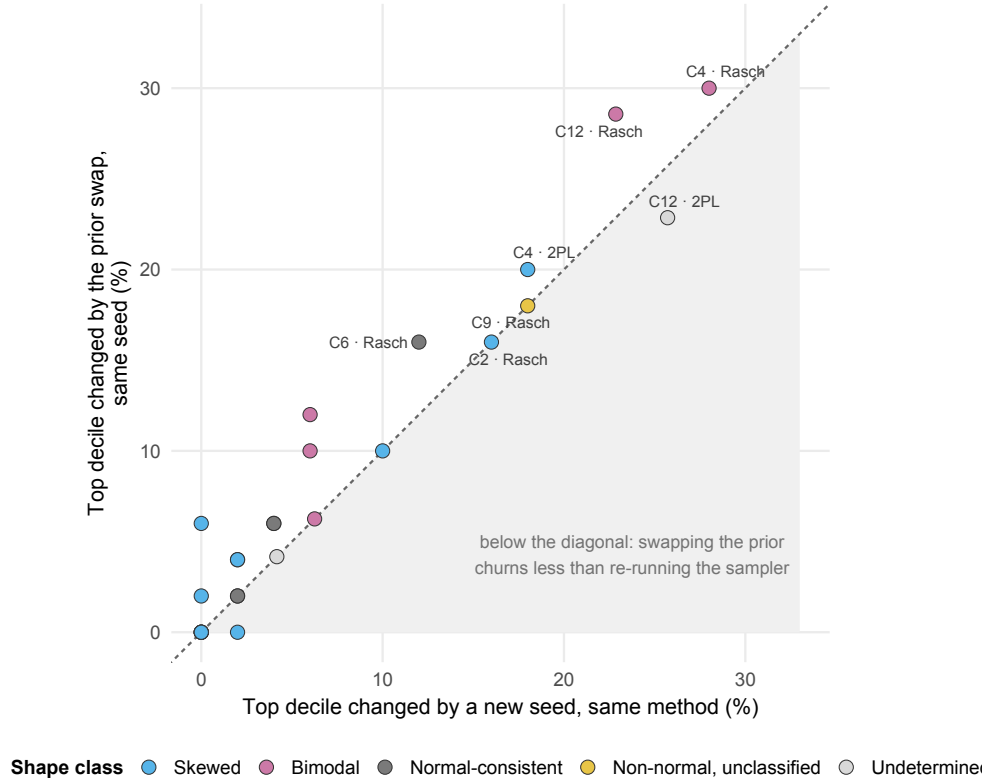


Note. Rows are case-cells, columns cross the four consequence families with the two levers (prior swap at fixed summary; summary swap at fixed prior); color encodes the observed change as a multiple of the frozen materiality threshold, and black dots mark case-cells whose materiality flag is set under that family's full rule, which considers more than the plotted statistic. The tails-and-cuts columns are the case-study cut-share family, measured directly on the reported estimates; they are distinct from the simulation's floor-bound tail loss. KS = Kolmogorov-Smirnov distance.

For rankings, the correct benchmark is the churn of the method against itself. [Figure F.3](#) plots each cell's between-prior top-decile churn against the churn of an identical re-run at a new seed: only the prior-family swap on bimodal cells (10.0%

against a 6.2% seed floor) clearly exceeds its benchmark, elicitation swaps sit at the floor, and several cells fall below the identity line, meaning the modeling choice moves fewer selections than the seed does. The worst single-cell seed churn replaces about a quarter of a top decile (Gaussian+PM subset maximum 28%), a floor that any selection-stability claim for these short tests must respect. Short Rasch forms are additionally banded: with a 0.02-SD gap rule, the 12-to-16-item forms yield a median of 14 distinct estimate bands, the largest holding 16% of the sample, which is what makes cut shares fragile when a band sits at a cut (the 19.0 percentage-point C4 episode; a 10.6-point episode at C2 is the only other exceedance).

FIGURE F.3. Between-Prior Selection Churn Against the Same-Method Seed Floor

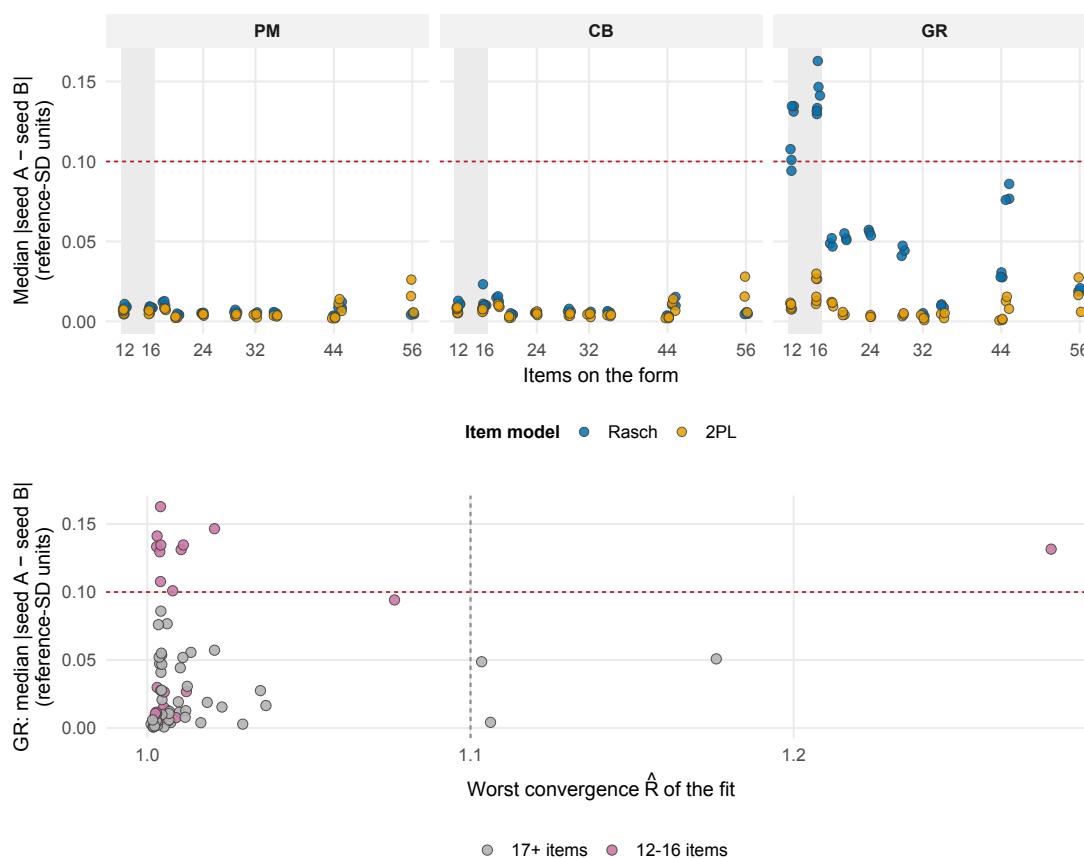


Note. Each point is one case-cell; the horizontal axis gives top-decile churn between two seeds of the identical pipeline and the vertical axis churn between priors at a common seed. Points below the identity line change fewer selections than the seed does.

F.4. Reproducibility of the Summaries. Across all 78 seed-A/B fit pairs, PM reproduced within 0.026 SD and CB within 0.028 SD, including the four worst- $\hat{R}$ fits.

GR exceeded the 0.10-SD self-reproduction bar on 11 of 78 fits, all Rasch forms of 12 to 16 items (11 of 24 short-form fits; 0 of 54 with 17 or more items), with a worst movement of 0.163 SD and a worst between-seed correlation of .938. Ten of the eleven failures had maximum $\hat{R}$ below 1.05, so chain diagnostics do not catch the problem; the mechanism audit shows sorted GR ensembles moving by only 0.001 to 0.005 SD while 328 to 493 persons change rank, so a nearly fixed quantile ensemble is being reassigned across dense near-rank ties. Figure F.4 displays the pattern of failures. The operational rule in Section 8 follows.

FIGURE F.4. Seed-to-Seed Movement of the Three Summaries by Test Length



Note. Median seed-to-seed movement of reported estimates per fit, by summary, against test length and against maximum $\hat{R}$; the 0.10-SD bar is the registered self-reproduction criterion. GR failures concentrate on 12-to-16-item Rasch forms and are invisible to $\hat{R}$. PM = posterior mean; CB = constrained Bayes; GR = triple-goal.

TABLE F.1. Simulation Verdicts Transported to the 19 Eligible Case-Cells

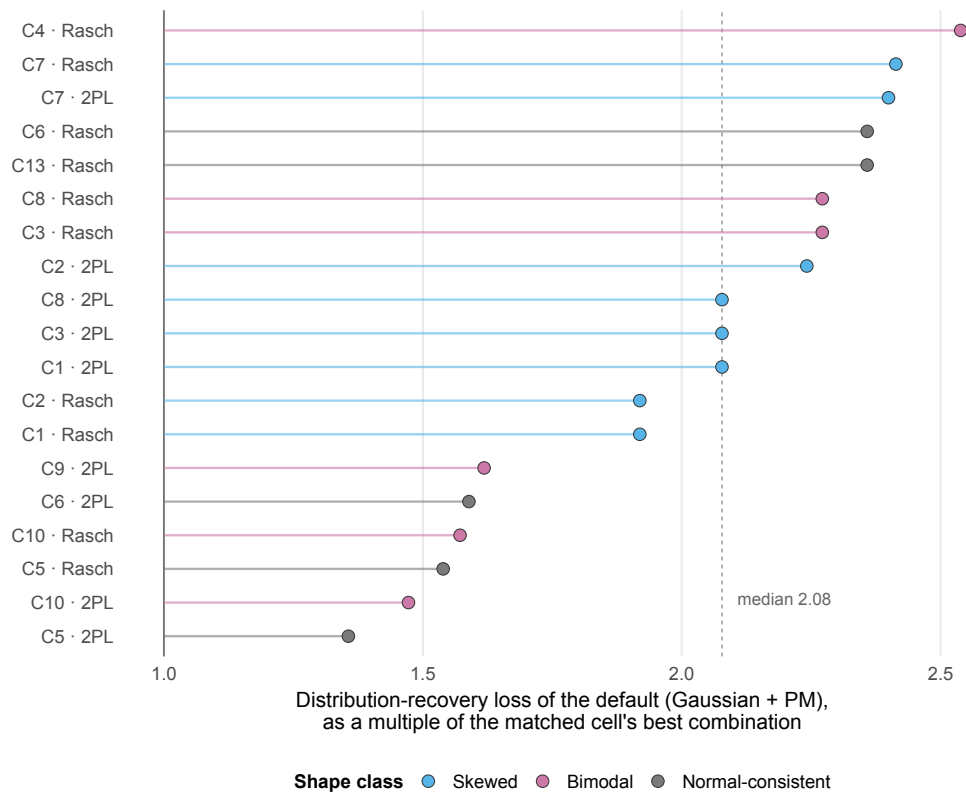
Case-cell	Class	Matched cell ($N, \bar{\rho}$)	KS ratio	Label	Winner	Default cost
C4-Rasch	bimodal	500, .8	0.48	strong win	DP+GR	2.54
C7-Rasch	skewed	500, .7	0.70	strong win	DP+GR	2.41
C7-2PL	skewed	500, .8	0.48	strong win	DP+GR	2.40
C3-Rasch	bimodal	500, .9	0.48	strong win	DP+GR	2.27
C8-Rasch	bimodal	500, .9	0.48	strong win	DP+GR	2.27
C2-2PL	skewed	500, .7	0.53	strong win	DP+GR	2.24
C1-2PL	skewed	500, .9	0.51	strong win	DP+GR	2.08
C3-2PL	skewed	500, .9	0.51	strong win	DP+GR	2.08
C8-2PL	skewed	500, .9	0.51	strong win	DP+GR	2.08
C1-Rasch	skewed	500, .8	0.72	strong win	DP+GR	1.92
C2-Rasch	skewed	500, .8	0.72	strong win	DP+GR	1.92
C9-2PL	bimodal	500, .7	0.71	strong win	DP+GR	1.62
C10-Rasch	bimodal	200, .8	0.76	strong win	DP+GR	1.57
C10-2PL	bimodal	200, .8	0.73	strong win	DP+CB	1.47
C13-Rasch	normal	500, .7	1.04	flag	G+GR	2.36
C6-Rasch	normal	500, .7	1.04	flag	G+GR	2.36
C6-2PL	normal	500, .7	0.99	flag	DP+GR	1.59
C5-Rasch	normal	500, .9	1.03	flag	G+GR	1.54
C5-2PL	normal	500, .9	1.01	flag	G+GR	1.36

Note. Each eligible case-cell is matched to its nearest simulation condition by fitted model, sample size, reliability tier, and shape class (median tier gap .033, maximum .0495). KS ratio is the matched condition’s mean distribution-recovery loss for DP(focused)+GR relative to Gaussian+GR; labels are the simulation’s registered evidence labels, whose residual flag label on the five normal-control rows (ratios .99–1.04) we read as parity with imprecision rather than as harm. Winner is the distribution-goal winner among the six prior-summary combinations at that cell; the individual-score winner is a PM combination in all 19 rows. Default cost is the factor by which the conventional default (Gaussian+PM) exceeds the winner’s distribution-recovery loss at the matched cell; against the pre-specified Gaussian+GR comparator instead, the median cost is 1.20. The join of case coordinates to simulation cells was assembled after both studies’ results were known and is exploratory (Section 7). DP = Dirichlet process (focused); G = Gaussian; 2PL = two-parameter logistic model; PM = posterior mean; CB = constrained Bayes; GR = triple-goal; KS = Kolmogorov–Smirnov.

F.5. The Verdict Join. The join underlying Table F.1 matches each eligible case-cell to its nearest simulation cell on four coordinates (model family, sample size matched downward, reliability tier, shape class), with negative-skew cells matched to the positive-skew condition by reflection, exact for symmetric quantities. Rows below the grid’s reliability support (C12-Rasch at $\bar{\rho} = .42$; C4-2PL at .26) are excluded rather than clamped; the withheld-cells display in the companion materials shows the simulation’s safety corner beside C12’s own movement without joining

them. Figure F.5 shows the cost of the default per row against the selected winner (median 2.08); against the prespecified Gaussian+GR comparator, which isolates the summary swap, the medians are 1.20 overall and 1.15 on the non-normal rows. Because the join was assembled after both studies' results were known, every number it produces is exploratory; its frozen inputs (the coordinates, the simulation's condition-mean store, and the label rules) are nevertheless all registered objects, and the five normal-control rows landing at ratios of .99 to 1.04 is the outcome expected of a sound lookup.

FIGURE F.5. The Cost of the Conventional Default at the Matched Cells



Note. For each eligible case-cell, the factor by which Gaussian+PM exceeds the selected winner's distribution-recovery loss at the matched simulation cell; rows are ordered by the plotted ratio. The winner is chosen and scored in the same cell, so these are post-selected benchmarks, optimistic by construction; the prespecified Gaussian+GR comparator gives medians of 1.20 overall and 1.15 on the non-normal rows. PM = posterior mean; GR = triple-goal.

Appendix G. A Worked Example With the Software Stack

We walk one case study, the vocabulary checklist C4 (16 dichotomous items; 2,495 respondents in the warehouse; analyzed at $n = 500$; bimodal placement under Rasch at $\bar{\rho} = .77$), through the full pipeline: a synthetic rehearsal with `IRTsimrel`, concentration elicitation with `DPprior`, fitting and summarizing with `DPMirt`, and the duplicate-seed check that any GR report from a 16-item Rasch form requires. All three packages install from GitHub (`remotes::install_github("joonho112/<name>")`); the code below was run under R 4.6.0, and the printed values are actual outputs.

G.1. A Rehearsal on Synthetic Data. Before fitting a real data set whose truth is unknown, we first confirm that the intended pipeline can recover a distribution like the one suspected, at the design actually available. `IRTsimrel` makes the rehearsal one call, because it solves the inverse design problem directly: asking for a 16-item Rasch design at a target average-information reliability of .80 under a bimodal population returns the calibrated global discrimination multiplier $c^* = 1.280$ (absolute calibration error 1.5×10^{-7}),

```
library(DPMirt) # wraps IRTsimrel for simulation
sim <- dpmirt_simulate(n_persons = 500, n_items = 16, model = "rasch",
                      target_rho = 0.80, latent_shape = "bimodal",
                      seed = 42)
```

and `sim$response` is a 500×16 matrix whose generating traits `sim$theta` are known, so the whole of what follows can be rehearsed with the truth in hand before the real matrix is touched. (The rehearsal’s target metric is the average-information coefficient; it exceeds the stricter [Equation \(3\)](#) slightly by Jensen’s inequality, as noted in [Section 2.2](#).)

G.2. Eliciting the Concentration Hyperprior. The focused arm’s rule ([Appendix C](#)) sets the cluster-count target at $\mu_K = 25$ for $n = 500$ with medium confidence, and `DPprior` converts that belief into a Gamma hyperprior exactly:

```
library(DPprior)
elic <- DPprior_fit(J = 500, mu_K = 25, confidence = "medium")
print(elic)
#> Gamma Hyperprior: alpha ~ Gamma(a = 8.904, b = 1.626)
#> Target (J = 500): E[K_J] = 25.00, Var(K_J) = 60.00
#> Achieved: E[K_J] = 25.00, Var(K_J) = 60.00
#> Method: A2-MN (5 iterations)
prob_w1_exceeds(0.5, elic$a, elic$b) #> 0.042
prob_w1_exceeds(0.9, elic$a, elic$b) #> 0.0004
```

The two closing lines are the weight diagnostics that the framework treats as required reporting: under this prior the first stick-breaking weight, the asymptotic mass of the cluster containing a randomly chosen respondent, exceeds one half with probability .042, so the cluster-count elicitation has not induced an effectively near-parametric prior (the unintended-prior check of [Appendix C](#)).

G.3. Fitting and Summarizing the Real Data. With the response matrix `resp` assembled from the warehouse (case `gcbs_brotherton_2013_vcl`: download the response table from the IRW, reshape to persons in rows and items in columns as a 0/1 matrix, and draw the 500-person analysis sample with a recorded seed; see the Data availability statement), the comparison we recommend is two fits and one line of summaries:

```
fit_g  <- dpmirt(resp, model = "rasch", prior = "normal",
                nchains = 4, niter = 8000, nburnin = 4000,
                thin = 2, seed = 101)
fit_dp <- dpmirt(resp, model = "rasch", prior = "dpm",
                alpha_prior = elic, nchains = 4, niter = 12000,
                nburnin = 4000, thin = 2, seed = 101)

diag <- dpmirt_diagnostics(fit_dp)
diag$n_clusters_summary; diag$ess_min_theta

est_g  <- dpmirt_estimates(fit_g, methods = c("pm","cb","gr"), seed = 1)
est_dp <- dpmirt_estimates(fit_dp, methods = c("pm","cb","gr"), seed = 1)
plot(fit_dp, type = "dp_density")      # posterior latent density + band
```

Each `est_*$theta` data frame holds the three reporting ensembles side by side. For C4 these six ensembles (plus the broad arm's three) are the nine reports displayed in [Figure 7](#): swapping the prior at fixed PM moves the reported scores visibly (KS distance .19 from the default report), and swapping the summary changes the family of the reported distribution, with the GR reports widening the ensemble and lengthening its lower tail relative to the default. At this case's matched simulation cell the transported verdict ([Table F.1](#)), an exploratory post-hoc lookup, is a strong win for DP+GR on the distribution goal, with the conventional default carrying 2.54 times the winner's distribution-recovery loss at that cell (a post-selected benchmark; the prespecified Gaussian+GR comparator gives 1.23), and PM remaining the winner for individual scores.

G.4. The Duplicate-Seed Check. C4 is a 16-item Rasch form, inside the region where GR person estimates can fail to reproduce themselves across seeds ([Section 7.3](#)). The check is a re-run and one summary:

```
fit_dp2 <- dpmirt(resp, model = "rasch", prior = "dpm",
                 alpha_prior = elic, nchains = 4, niter = 12000,
                 nburnin = 4000, thin = 2, seed = 202)
est_dp2 <- dpmirt_estimates(fit_dp2, methods = "gr", seed = 1)
sd_ref <- sd(est_g$theta$pm) # common Gaussian-PM reference SD
median(abs(est_dp$theta$gr - est_dp2$theta$gr)) / sd_ref
```

The statistic is the median absolute per-person movement between the two runs' aligned GR ensembles, scaled by the common Gaussian-PM reference SD, which is the diagnostic that produced the case-study counts. If it exceeds about a tenth of the reference SD, the GR ensemble should not be reported from either single run; report CB in its place, or report GR only once two independent seeds agree, and say so. In the case studies this check failed on 11 of 24 short-form fits, every failure a Rasch fit, while every PM and CB ensemble reproduced within 0.03 SD, and nothing in the chain diagnostics distinguished the failures, which is why the check is part of the recommended workflow rather than an optional step.